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MILITARY SPECIFICATION

NAVIGATIONAL SET, DOPPLER AN/ASN-128( ) (XE-2)

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of a Doppler Navigation System (DNS), and is designated AN/ASN-128( ). The AN/ASN-128( ) consists of the following units:

a. Receiver - Transmitter - Antenna, Radar RT-1193/ASN-128( )

b. Converter, Signal Data, Radar CV-3338/ASN-128( )

c. Computer - Display Unit, CP-1252/ASN-128( )

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the Department of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Cdr, U.S. Army Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, NJ 07703-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-C-172	Cases; Bases, Mounting; Mounts, Vibration, for use with Electronic Equipment in Aircraft.
MIL-T-713	Twine: Impregnated, Lacing and Tying.
MIL-B-5087	Bonding, Electrical, and Lightning Protection for Aerospace Systems.
MIL-E-5400	Electronic Equipment, Airborne General Specification for.
MIL-C-6781	Control Panel, Aircraft Equipment Rack or Console Mounted.
MIL-M-7793	Meter, Time Totalizing.
MIL-P-11268	Parts, Materials and Processes Used in Electronic Communication Equipment.
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.
MIL-T-43435	Tape, Lacing and Tying.
MIL-P-55110	Military Specifications, Printed Wiring Board.

STANDARDS

MIL-L-85762	Lighting AN/AVS-6 Aviators Night Vision Imaging System (ANVIS) Compatible.

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-461	Electromagnetic Interference Characteristics Requirement for Equipment.
MIL-STD-462	Electromagnetic Interference Characteristics Measurements of
MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of.
MIL-STD-781	Reliability Tests, Exponential Distribution.
MIL-STD-810	Environmental Test Methods.
MIL-STD-883	Test Methods and Procedures for Microelectronics.

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MIL-STD-1130	Connection, Electrical, Solderless Wrapped.
MIL-STD-1472	Human Engineering Design Criteria.
MIL-STD-1562	List of Standard Microcircuits.
MIL-STD-1686	Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices) (Metric).

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5904).

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

INTERNATIONAL STANDARDS, AF

ASCC

AIR STD 53/13( )	The Specification for Evaluation of The Accuracy of Airborne Doppler Navigation Systems
AIR STD 53/14	Definition of Sea State for Use in Connection with Doppler Navigation

(Copies are available from: Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

AVRADA

AV-IT58001	Installation and Acceptance Testing of Navigational Set, Doppler AN/ASN-128 in Army Aircraft

DRAWINGS

USACECOM

DL-SM-B-858489	Receiver-Transmitter-Antenna, Radar, RT-1193/ASN-128( ).
DL-SM-B-858425	Converter, Signal Data, Radar CV-3338/ASN-128( ).
DL-SM-B-858970	Computer-Display Unit, CP-1252/ASN-128( ).
SM-A-858393	Switch, waveguide.
SM-A-858394	Receiver-Transmitter module.

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SM-A-858395 Generator, Reference Signal.

(Copies required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both title and number or symbol should be stipulated when requesting copies).

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The Doppler Navigation System (DNS) provides position data in the Universal Transverse Mercator (UTM) coordinate system and also in Latitude/Longitude; the operator can select either coordinate system for displaying present position, destinations and targets-of-opportunity. The operator can also select for display Groundspeed and Track Angle, Distance-Off-Course (XTK), and Track Angle Error (TKE), Distance, Bearing and Time-to-go to 10 destinations, Wind Speed and Direction (if true air speed is available) and UTM Spheroid and Magnetic Variation. Position, velocity and steering data are also provided in digital form for use in systems and equipment external to the DNS.

3.1.1 Government-furnished equipment (see 3.21.1 and 6.6). The Government shall furnish the following property for use in each DNS on order:

NSN No.	Nomenclature	Type	Quantity
6135-00-835-1023	Battery	BA-1090/U	1 ea

3.2 Construction. The three units comprising the DNS shall be constructed in accordance with the drawing and data lists indicated below.

DL-SM-B-858489	Receiver-Transmitter-Antenna, Radar, RT-1193/ASN-128( ).
DL-SM-B-858425	Converter, Signal Data, Radar CV-3338/ASN-128( ).
DL-SM-B-858970	Computer-Display Unit, CP-1252/ASN-128( ).

3.3 First article samples (see 4.3 and 6.2). When specified, the contractor shall furnish sample unit(s) for first article inspection and approval.

3.4 Cleaning (see 4.13).

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3.4.1 Parts. Prior to the application of conformal coating, on those parts affected, and after fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Post cleaning assembly processes shall minimize handling by bare hands to prevent contamination by fungus nutrients. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, and other parts.

3.4.2 Units. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.5 Weight (see 4.15). The weight of each completely assembled unit shall not exceed the following:

a.	Receiver-Transmitter-Antenna, Radar RT-1193/ASN-128( ).	11.0 lb
b.	Converter, Signal Data, Radar CV-3338/ASN-128( ).	13.0 lb
c.	Computer-Display Unit, CP-1252/ASN-128( ).	7.5 lb

3.6 Controls (see 4.13). All movable controls shall operate properly without binding or other undue restriction. Controls shall not be assembled in a misaligned condition.

3.7 Finish, protective (see 4.4). The units shall be given protective finish in accordance with MIL-F-14072 and MIL-B-5087. This includes finish of hardware, such as screws, etc., and necessary touch-up after mounting. The final paint film on type I surfaces shall be film B in accordance with MIL-F-14072. The mating surfaces of the chassis, case and mounting shall be free of all insulating finishes in order to provide a continuous electrical bond between these items as specified in MIL-C-172 and to enable the installing activity to accomplish the 2.5 milliohm requirement of MIL-B-5087, (Class R Bonding) to the basic aircraft structure.

3.8 Securing of parts (see 4.13).

3.8.1 General. Brackets, lugs, inserts, bolts and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.8.2 Securing of parts by threaded fasteners. Nylok or other screws and nuts with plastic devices and Loctite or similar sealants shall not be subjected to temperatures in excess of 250°F (as during baking of paint). Loctite or similar sealants shall be applied in accordance with the manufacturer's instructions.

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3.9 Marking (see 4.4).

3.9.1 General. Marking shall conform to MIL-M-13231. Front panel marking shall be in accordance with group I of that specification and the equipment drawings.

3.9.2 Visibility. Wherever practicable, subassemblies (modules) shall be so mounted that their identification markings will be readily visible with minimum disassembly of the units.

3.9.3 Serial numbers. The units listed in 1.1 shall be serial numbered.

3.10 Soldering (see 4.13). Soldering shall conform to requirement 5 of MIL-STD-454 and the equipment drawings.

3.11 Wiring, cabling, bonds and grounds (see 4.13)

3.11.1 Cabling. Insulated wires shall be formed into cables, except where operation of the unit(s) would be adversely affected or where it is physically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance. The cabling of wires shall be effected by twine conforming to MIL-T-713 and tape conforming to MIL-T-43435. Individual conductors thus secured shall lie essentially parallel; however, this does not prohibit the use of twisted pairs.

3.11.2 Slack. Wires and cables shall be as short as practical, except that sufficient slack shall be provided for the following purposes:

a. To prevent undue stress on cable forms, wires, and connections including connections to resiliently supported parts.

b. To enable parts to be removed and replaced during servicing without disconnecting other parts.

3.11.3 Protection. Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges.

3.11.4 Support. Wire and cable shall be properly supported and secured to prevent undue stress on the conductors and terminals and undue change in position of the wire and cable:

a. During and after subjection of the unit(s) to specified service conditions, or

b. After service or repair of the unit(s) in a normal manner.

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3.11.5 Clearance. Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be such that no accidental contract can occur between adjacent connections when subjected to specified service conditions.

3.11.6 Splicing. Wires in a continuous run between two (2) terminals shall not be spliced during the wiring operation.

3.11.7 Connections, general. Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practical, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections; however, conformal coating is acceptable. Connections made by wire wrap in accordance with MIL-STD-1130 shall also be acceptable.

3.11.8 Bonds and grounds. Bonding for equipment current path returns and shock hazards shall be so installed that expansion, contraction, or movement incident to normal service use and maintenance will not break or loosen the connection. Surface preparation for bonds and grounds shall be accomplished by removing all anodic film, grease, paint, lacquer or other high-resistance coatings from the immediate areas of contact. The equipment finish shall be restored to its original condition in accordance with MIL-F-14072. The dc resistance of all bonds and grounds shall not exceed 2.5 milliohms.

3.11.9 Cable and connectors. All shielded cables shall have the shield terminated in an approved military connector with conductive finish per MIL-F-14072 and EMI back shell to provide peripheral bonding of the shield. Shielded cables shall be kept to the minimum practicable by use of alternative suppression and decoupling techniques such as pin filter connectors. All connectors and panel-mounted components and the panel mating surfaces shall be free of non-conductive finishes and provide positive bonding and grounding with mating connectors and equipment ground plane. The bonding resistance for all connectors and mounted components shall not exceed 2.5 milliohms.

3.12 Parts, materials and processes (see 4.4). Electronic components, parts, devices, materials, techniques and technologies, including microelectronics shall conform to MIL-P-11268 and MIL-STD-1686. Microcircuits shall be selected from MIL-STD-1562.

3.12.1 Elapsed time indicators (see 4.16). Electrochemical elapsed time indicators shall not be used. Digital indicators shall be used having a range of 1000 hours and a resolution of 1 hour. The ETI shall comply with the quality levels of MIL-M-7793.

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3.13 Service conditions (see 4.8). The unit(s) on order shall meet the service conditions as specified in MIL-E-5400 for class 1A equipment, except as specified, and meet the requirements as specified herein, after a warm-up period not to exceed five (5) minutes.

3.13.1 Temperature-altitude (see 4.8.1). The unit(s) on order shall be capable of meeting the temperature-altitude requirements through a temperature and altitude range of -62°C to +85°C from sea level to 30,000 feet. In addition temperature altitude tests shall be conducted at 23°C ± 10°C and 40,000 feet, nonoperating. The unit(s) shall be operable and conforming to limits specified from -54°C to +55°C and intermittent operation from -55°C to +71°C from sea level to 30,000 feet. The input supply voltages shall be from +22.5 to +30.0 Vdc.

3.13.2 Humidity (see 4.8.2). The unit(s) shall be capable of withstanding up to 100 percent relative humidity, including conditions wherein condensation takes place in the form of water. The unit(s) shall be operable without degradation in specified performance after exposure to the above conditions in a nonoperating condition.

3.13.3 Dust (see 4.8.3). The unit(s) shall be capable of withstanding, in both operating and nonoperating condition, exposure to dust particles.

3.13.4 Vibration (see 4.8.4). The DNS unit(s) normally mounted without the use of additional supporting pins or brackets shall be capable of withstanding vibration as shown in figure 2, curve IIIb, sheet 3 of MIL-E-5400 truncated at 500 Hz. The units shall operate without degradation in specified performance and shall sustain no physical damage, during and after exposure to the above conditions.

3.13.5 Shock (see 4.8.5).

3.13.5.1 Bench handling (see 4.8.5.1). The unit(s) shall be capable of meeting specified performance following bench handling.

3.13.5.2 Equipment (see 4.8.5.2). The unit(s) shall not suffer damage or subsequently fail to provide specified performance when subjected to 18 impact shocks of 15g, consisting of 3 shocks in opposite directions along each of 3 mutually perpendicular axes; each shock impulse shall have a time duration of 11 ± 1 milliseconds. The "g" value shall be within ±10 percent when measured with a 0.2 to 250 Hz filter, and maximum "g" value shall occur at approximately 5 1/2 milliseconds.

3.13.5.3 Crash safety (see 4.8.5.3). With excursion stops or bumpers in place and with maximum rated load applied in a normal manner, the mounting base, individual isolators or other attaching devices shall be capable of withstanding 12 impact shock of 30g, consisting of 2 shocks in opposite directions along each of 3

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mutually perpendicular axes. Each shock impulse shall have a time duration of 11 ± 1 milliseconds. The "g" value shall be within ±10 percent when measured with a 0.2 to 250 Hz filter, a maximum "g" value shall occur at approximately 5 1/2 milliseconds. Bending and distortion is permitted; however, there shall be no failure of the attaching joints. The equipment or dummy load (6.5) shall remain in place.

3.13.6 Fungus (see 4.8.6). The unit(s) in the assembled and ready for delivery condition shall provide no nutrients in materials, coating or contaminant form that support fungal growth.

3.13.7 Salt fog (see 4.8.7). The unit(s) shall be resistant to the corrosive effects of a salt-sea atmosphere.

3.13.8 Explosive atmosphere (see 4.8.8). The unit(s) shall not cause ignition of an ambient-explosive-gaseous mixture with air when operating in such an atmosphere.

3.13.9 Acceleration (see 4.8.9). The unit(s) shall be capable of operating without degradation in specified performance, and shall sustain no physical damage, when exposed to steady-state acceleration levels of up to 6g for at least one (1) minute.

3.13.10 Rain (see 4.8.10). The unit(s) normally mounted and exposed to periods of heavy precipitation with intermittent wind, shall be capable of meeting specified performance and shall sustain no physical damage.

3.13.11 RTA seal (see 4.8.12). The receiver-transmitter antenna shall be sufficiently sealed against entry of contaminant fluids when tested, in accordance with 4.8.12, to pass the operational Self-Test, in accordance with 3.18.1.9, and not show a gain of more than 8 grams when weighed in step h of 4.8.12.

3.13.12 RTA immersion (see 4.8.13). The receiver-transmitter antenna shall show no evidence of leakage via the formation of air bubbles when tested in accordance with 4.8.13.

3.14 Reliability (see 4.6). The AN/ASN-128( ) equipment shall have a specified mean-time-between-failure (θ_o) of 1000 hours.

3.15 Electromagnetic interference (see 4.8.11). The emission of electromagnetic interference by the AN/ASN-128 and the susceptibility of the AN/ASN-128 to electromagnetic interference shall comply with the following requirements of MIL-STD-461 as specified herein.

3.15.1 Emission and susceptibility. The unit(s) shall meet the following emission and susceptibility requirements of MIL-STD-461, notice 4:

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CE01	CS01	RE02	RS02
CE03	CS02	RE02.1	RS03
CE04	CS06		
CE05			

a. The frequency limit for Requirement RS03 is from 10 KHz to 1 GHz, and from 12.4 GHz to 18 GHz, except for 13.325 GHz ± 80 MHz. The subsystem and any equipment item thereof shall be subjected to the following susceptibility field intensity levels:

10 KHz to 1.9 MHz	-	1 volt/meter
2 MHz to 1000 MHz	-	10 volts/meter
12.4 GHz to 18 GHz	-	1 volt/meter (except no test required from 13.245 to 13.405 GHz)

b. Conducted emissions on system interface lines shall meet the requirements of CE03 and CE05, except that any narrow-band energy (i.e., harmonics related to the frequency of the square wave of the basic transmissions) in the frequency band of 7.5 KHz to 600 KHz resulting from interface digital data signals shall be completely free of any limits.

3.16 Electrical, Receiver-Transmitter-Antenna, Radar, RT-1193/ASN-128( ) (see 4.9). Unless otherwise specified, the Receiver-Transmitter-Antenna (RTA) shall meet the requirements specified in 3.16.1 through 3.16.8 under service conditions as specified in paragraph 3.13 within five (5) minutes after the RTA has been turned on.

3.16.1 RTA beams (see 4.9.1). The RTA shall provide four (4) non-coplanar beams of electromagnetic energy.

3.16.1.1 Beam switching (see 4.9.1). The beams shall be generated sequentially at a rate not to exceed 7.5 Hz. The sensitivity shall be 20 dB minimum for each beam.

3.16.1.2 Spurious. The beam-to-beam noise unbalance shall not exceed 4 dB if any of the sensitivity readings are between 20 and 22 dB.

3.16.1.3 Beam Leakage. The leakage amplitude shall not be greater than 2.0 V peak-to-peak on each beam, and for all beams, it shall be 0.8 ± 0.7 V peak-to-peak.

3.16.2 Frequency (see 4.9.1.1). The radiated electromagnetic energy (RF) of the RTA shall be in the frequency range of 13.25 GHz to 13.40 GHz.

3.16.2.1 Modulation (see 4.9.1.2). The RTA RF shall be frequency modulated at 30 KHz by an externally generated signal whose modulation index shall be 0.92 ± 15%.

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3.16.3 Radiated power (see 4.9.1.3). The RF transmitter power output to each beam shall not exceed 100 milliwatts.

3.16.3.1 Receiver Image. The receiver image signal ratio shall be 20 decibles below or more.

3.16.4 Velocity limits (see 4.9.1.4). The maximum RTA velocity limits when operated with the Signal Data Converter (3.17) shall be as follows:

Along Vehicle Axis Velocity (Vx)	-50 to +350 knots (-93 to +650 km/hr)
Across Vehicle Axis Velocity (Vy)	-100 to +100 knots (-185 to +185 km/hr)
Vertical Vehicle Velocity (Vz)	±5000 ft/minute (±1500 meters/minute)

3.16.5 Velocity accuracy (see 4.10.7). The DNS (1 SIGMA) velocity accuracy over the full range (see 3.16.1 - 3.16.4 and 3.16.6 - 3.16.8, 3.17, and 3.18) shall be as follows:

	LAND	SEA*
Vx	0.25% Vt + 0.1 Kt (0.185 km/hr)	0.35% Vt + 0.1 Kt (0.185 km/hr)
Vy	0.25% Vt + 0.1 Kt (0.185 km/hr)	0.35% Vt + 0.1 Kt (0.185 km/hr)
Vz	0.1% Vt + 0.05 Kt (0.093 km/hr)	0.1% Vt + 0.05 Kt (0.093 km/hr)

* Notes:

1. $Vt = \sqrt{Vx^2 + Vy^2 + Vz^2}$
2. Exclusive of water motion

The DNS shall be tested to assure that the circuits of the transmitter, receiver leakage elimination filter, frequency tracker, timer, power supply, and navigation computer are functioning correctly and velocity accuracy requirements are met.

3.16.5.1 Navigation accuracy tests (see 4.10.7). Flight tests shall be conducted in accordance with an approved flight test plan to demonstrate the navigation performance of the Doppler Navigation Set. The following performance criteria shall be met over land:

Heading Reference Accuracy (°RMS)	Along Track Error (% distance traveled)	Cross Track Error (% distance traveled)
Inertial 0.5	0.35 (1σ)	0.94 (1σ)
Magnetic 1.0	0.35 (1σ)	1.78 (1σ)

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The along track errors listed in the above table are the Doppler errors. The cross track errors listed in the above table are a combination of Doppler plus the heading reference (the dominant factor) errors.

During flight test the present position is also affected by additional errors due to map errors, parallax error during position updates, and position quantization.

The above requirement assumes a properly aligned heading reference is used in conjunction with ASN-128 DNS. Visual position fixing errors of the aircraft over any test range checkpoint shall not exceed 10 meters. The total error in the knowledge of the checkpoint coordinates used shall not exceed two (2) meters for engineering or production flight tests. Each leg shall be a minimum of 20 kilometers long.

3.16.6 Altitude limits (see 4.9.1.5 and 4.10.8). (Above terrain) The RTA operational altitude limits when operated with the Signal Data Converter (3.17) shall be as follows:

Terrain	Altitude
Land	0 to 10,000 ft (0 to 3,000 meters)
Smooth sea (B1) (WMO 2) per AIR STD 53/14	0 to 10,000 ft (0 to 3,000 meters)

3.16.7 Altitude limits (see 4.9.1.5, 4.10.1 and 4.10.8). (to 10,000 ft altitude) The RTA operational attitude limits when operated with the Signal Data Converter (3.17) shall be as follows:

Terrain	Altitude	
	Pitch	Roll
Land	±30 deg	±45 deg
Smooth sea (B1) (WMO 2) per AIR STD 53/14	±20 deg	±30 deg

3.16.8 RTA input/output interface (see 4.9.1.6). The RTA shall functionally interface with the Signal Data Converter (3.17). The interface parameters shall consist of the following:

Input	Function
7.5 Hz square wave	Beam switching (B2 + B3)
7.5 Hz square wave	Beam switching (B3 + B4)
30 kHz square wave	FM modulation frequency
+5 V, ±15 Vdc	Primary dc source for RF power generation and RTA electronics
+5V discrete	Set RTA in Test Mode

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Output	Function
IF output	4 beam frequencies
Six calibration signals	Antenna calibration Constants (six constants are transmitted)

3.16.9 RF components. The parameter of the RF components in the DNS is as specified in the SM-A-858393, SM-A-858394 and SM-A-858395 drawings.

3.17 Converter, Signal Data, Radar CV-3338/ASN-128( ) (see 4.10). The Signal Data Converter (SDC) shall meet the requirements specified in 3.17.1 under the service conditions as specified in 3.13 with the equipment oriented in any position, within five (5) minutes after the equipment has been turned on.

3.17.1 Analog signal inputs (see 4.10.1).

3.17.1.1 Heading, roll and pitch. The SDC shall be capable of receiving 26/11.8 Vac 400 Hz control transmitter signals at an angular rate not to exceed 50 degrees/second for heading, roll and pitch of the following characteristics. These synchro input signals shall be at one degree of shaft rotation per degree of aircraft heading and attitude change.

3.17.1.1.1 Input accuracy (one SIGMA values).

Parameter	Accuracy
Roll	3.0°
Pitch	3.0°
Heading	
(Magnetic)	1.0°
(Inertial/Gyro Compass)	0.5°

3.17.1.1.2 Synchro loading. The SDC line-to-line load on all synchro inputs shall be 5000 ohms or greater.

3.17.1.1.3 Conversion accuracy (see 4.10.1). The SDC heading, roll and pitch synchro signal conversion accuracy shall be ±0.45 degrees (3 SIGMA) when the synchro output impedance (Zss) is less than or equal to 100 +j30 ohms for heading and 150 +j60 ohms for roll and pitch.

3.17.1.2 SDC on/off discrete (see 4.10.2). Prime power to the SDC shall be controlled by receipt of a discrete. This discrete shall be +5 ± 0.5 Vdc to turn the SDC "ON" and 0 ± 0.5 Vdc to turn the SDC "OFF".

3.17.1.2.1 Circuitry. The SDC ON/OFF receiving circuit shall have the following electrical characteristics:

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Resistance	-	12,000 ohms minimum
Capacitance	-	100 picofarads maximum

3.17.1.2.2 Noise immunity (see 4.10.2.1). The SDC ON/OFF receiving circuit shall be immune to noise pulses having a time-voltage integral of 1 volt amplitude in any period of 50 microseconds. These noise pulses are in addition to the signal levels set forth in 3.17.1.2.

3.17.2 RTA, SDC radar initialization (see 4.10.3). Upon receipt of the ON/OFF discrete (3.17.1.2), the SDC shall output serial data words in the format shown in figure 1 (Test). The content of the data shall be as shown in table I, after which the SDC shall output navigation data.

3.17.2.1 Initialization period (see 4.10.3). The initialization period shall be 18 ± 2 seconds.

3.17.2.2 Signal acquisition (see 4.10.4). Automatic acquisition and tracking of a trackable Doppler radar signal shall occur in less than 10 seconds.

3.17.3 SDC digital output (see 4.10.5). The SDC digital data output shall consist of one two-wire circuit.

3.17.3.1 Electrical characteristics. The data word transmission shall use a bipolar signal with a return to zero logic.

3.17.3.1.1 Voltage level (see 4.10.5.1). All digital data output signals shall conform to the voltage levels shown in figure 2.

3.17.3.1.2 Circuitry (see 4.10.5.2). The SDC driver circuitry shall be of the balanced type shown in Figure 3. The output circuit shall be capable of accommodating any combination of the following load parameters.

Resistance	-	600 to 12,000 ohms
Capacitance	-	1,000 to 30,000 picofarads

3.17.3.1.3 Noise immunity (see 4.10.5.3). The SDC digital transmitter shall be immune to noise pulses having a time-voltage integral of 1 volt amplitude in any period of 50 microseconds. These pulses are in addition to the signal levels set forth in figure 2.

3.17.3.1.4 Reverse current (see 4.10.5.4). The driver circuit shall be capable of withstanding up to one milliampere of reverse current without damage to the line driver components or circuitry.

3.17.3.2 Digital data waveform (see 4.10.6). The digital data waveform shall be contained in a self-synchronizing waveform (figure 5). The waveform shall exhibit a voltage level transition at the

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beginning of each cycle or bit and return to zero in the middle of each cycle. The middle of the cycle shall be defined as being within ±10% of one half cycle as shown in figure 4.

3.17.3.3 Word structure (see 4.10.6.1). The standard word structure for the SDC serial data bus shall consist of 24 bits as shown in tables I and II. The maximum digital data transmission shall not exceed 93 words/second. The following language constants shall apply to tables I and II and to the data transmissions described therein.

3.17.3.3.1 Word/bit constants (see 4.10.6.2). Status bit definitions shall be as follows:

MEMORY (M)	(BIT 22)	-0 = MEMORY 1 = TRACK
TEST	(BIT 23)	-0 = NAVIGATE 1 = TEST
POWER SUPPLY FAIL (F)		-0 = NORMAL 1 = FAIL
TAS AVAILABLE		-0 = TAS not available 1 = TAS available

3.17.3.3.2 Word synchronization (see 4.10.6.3). A blank consisting of the synch voltage on both transmission wires for a minimum period of four cycles shall exist between successive words that are transmitted on the same transmission system. This gap, as shown in figure 5, shall provide the necessary synchronization for receiving equipment.

3.17.3.4 Data cycle and data rate (see 4.10.6.4). The SDC digital output data frame cycle and data rates shall be as shown in figure 1.

3.17.3.5 Bit synchronization (see 4.10.6). The clock or bit synchronization signal shall consist of a 7.6 kHz ± 100 Hz square wave signal having a rise and fall time of 10 ± 5 microseconds as shown in figure 4.

3.17.3.6 Velocity Test (SDC) (see 4.11.1.2). The SDC shall be tested to assure that the circuitry of the leakage elimination filter, frequency tracker, timer/interface, A/D converter, and the power supply are functioning correctly and accuracy requirements are met when inputs of attitude and velocity meet the limits and accuracy described in 3.16.4 through 3.17.1.1.3.

3.18 Electrical, Computer-Display Unit CP-1251/ASN-128( ) (see 4.11). Unless otherwise specified, the computer-display unit (CDU) shall meet the requirements specified in 3.18.1 through 3.20.2.6.2 under service conditions as specified in 3.13 within five (5) minutes after the sensor has been turned on, and the ANVIS (Aviators Night Vision Imaging System) lighting requirements as specified in MIL-L-85762.

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3.18.1 Functional capability. The CDU shall be programmed to enable the CDU to accept valid SDC serial binary words (as shown in tables I and II) and use the data contained therein to provide the following functional capability (see 4.11).

3.18.1.1 Area of operation (see 4.11.1). The CDU shall be capable of worldwide operation and shall provide for selection of aircraft position data in either Latitude-Longitude coordinates or the coordinates of the UTM military grid reference system. The CDU shall provide a navigation capability from 80 degrees North Latitude to 80 degrees South Latitude using the CLARK 1866, CLARK 1880, BESSEL, INTERNATIONAL, EVEREST (or MALAYAN) and AUSTRALIAN Spheroids. In that the MALAYAN Spheroid has not been defined, the EVEREST constants shall be used for MALAYAN Spheroid operation. During operation involving zone and area boundary crossing, the CDU present position shall not be degraded. The CDU shall be tested to assure that the circuits of the Central Processing Unit, RAM/ROM, I/O, Numeric Logic, and post regulator are functioning correctly and accuracy requirements are met when inputs of attitude and velocity meet the limits and accuracy described in 3.16.4 through 3.16.7 and 3.17 through 3.17.1.2.2.

3.18.1.2 Present position updating (see 4.11.2). No interruption of the normal UTM or LAT/LONG (navigation) mode shall occur during present position manual or automatic updating and there shall be no degradation in navigation accuracy.

3.18.1.2.1 Automatic updating (see 4.11.2.1). The CDU shall have the facility for automatic updating of present position with any selected, stored checkpoint. That is, it shall be possible to automatically update the CDU present position readout when a stored checkpoint is directly overflown, by using the coordinate of that checkpoint.

3.18.1.2.2 Manual updating (see 4.11.2.2). When a valid checkpoint is overflown, it shall be possible to manually update present position by entering the known coordinates individually.

3.18.1.3 Checkpoint (see 4.11.3). The CDU shall have the facility for entering and storing 10 checkpoints in either UTM or Latitude/Longitude.

3.18.1.4 Target storage (see 4.11.4). The CDU shall be capable of freezing and storing the present position coordinates of any target of opportunity overflown. Storage locations 6, 7, 8, and 9 shall be available for the storage of the coordinates.

3.18.1.5 Ground track-ground speed (see 4.11.5 and 4.11.15.1). The CDU shall compute and display ground track-ground speed information from Magnetic Variation, Heading and SDC velocity inputs. The CDU display resolution of track angle shall be $\pm$1 degree and that of ground speed shall be $\pm$1 km/hr.

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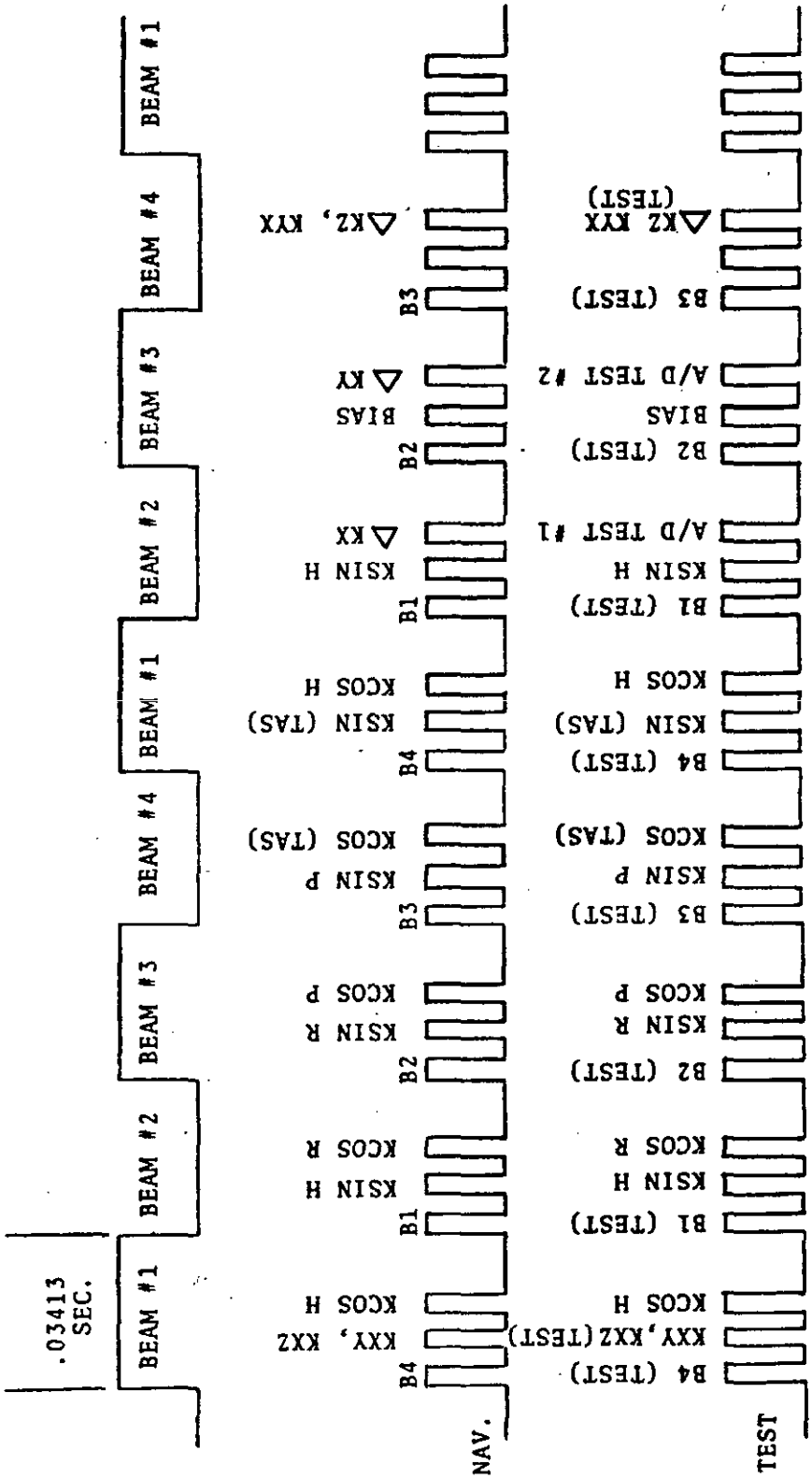
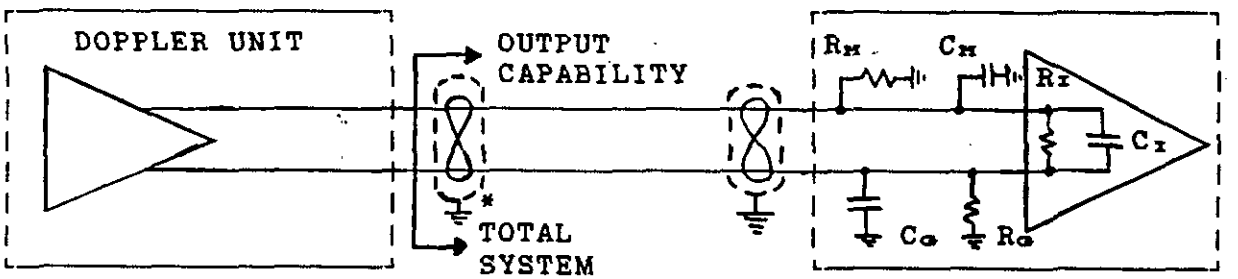


FIGURE 1. Data Transmission Cycle.

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Circuit	Transmitted voltage						Received voltage					
	One		Zero		Sync		One		Zero		Sync	
	Nom	Tol	Nom	Tol	Nom	Tol	Nom	Tol	Nom	Tol	Nom	Tol
Line A to Line B	+10	±1	-10	±1	0	±1	+10	±3	-10	±3	0	±3
Line A to Ground	+5	±0.5	-5	±0.5	0	±0.5	+5	±1.5	-5	±1.5	0	±1.5
Line B to Ground	-5	±0.5	+5	±0.5	0	±0.5	-5	±1.5	+5	±1.5	0	±1.5

Figure 2. Digital Signal Electrical Characteristics



OUTPUT (SYSTEM) CAPABILITY		UTILIZATION DEVICE STANDARDS
Total System** resistance	600 to 12,000 ohms	R_I = 12,000 ohms minimum
Total System** capacitance	1,000 to 30,000 picofarads	C_I = 100 picofarads maximum
System capacitance unbalance	Not defined but unbalance due to aircraft interwiring should be held to a minimum	R_M or $R_a \geq R_I$ C_M /C_a Unbalance $\geq$ 10 picofarads

NOTES

* Shields to be grounded in aircraft at both ends of all "breaks".

** Includes aircraft interwiring.

FIGURE 3. Input and output circuit standards.

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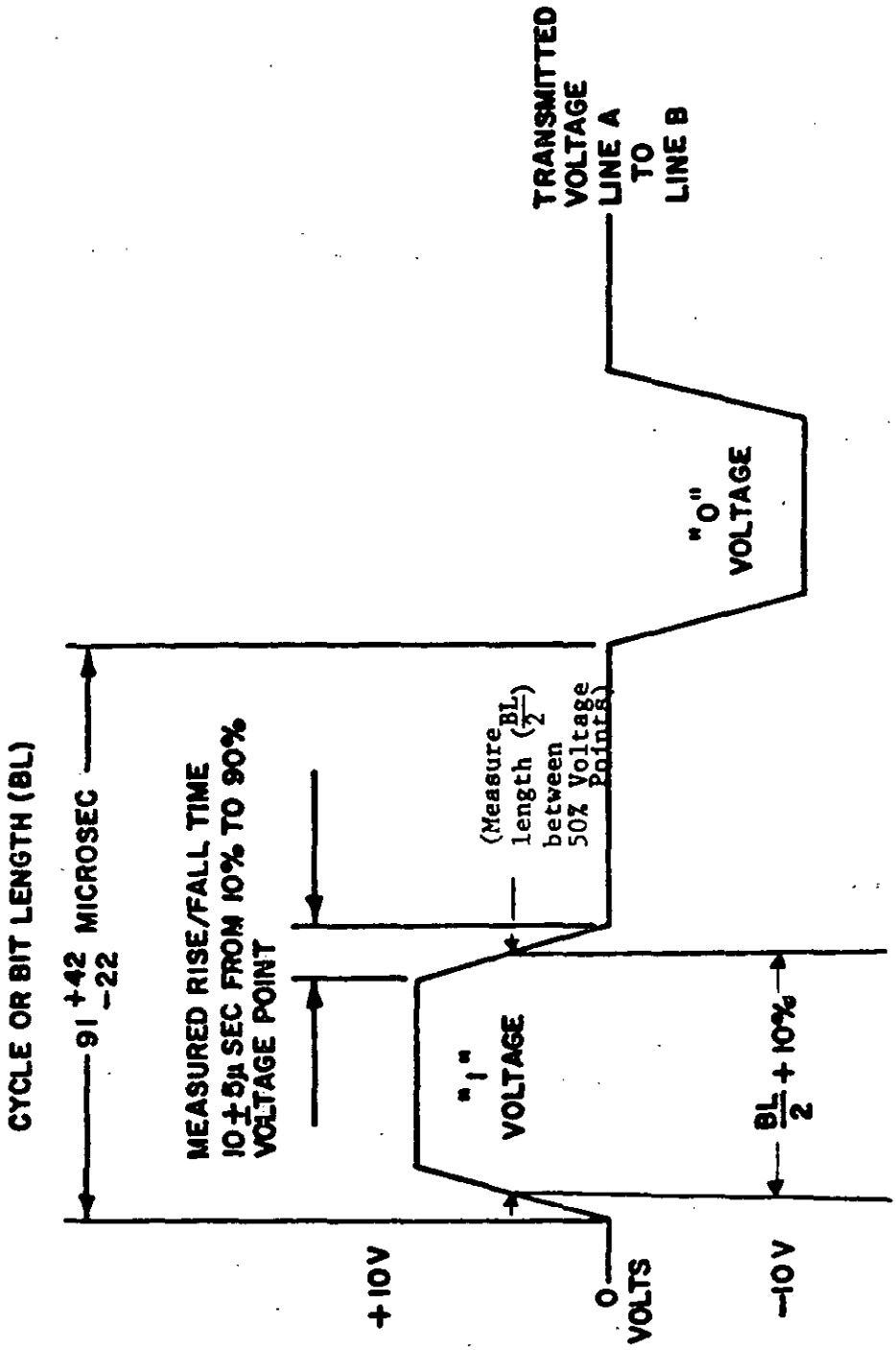


FIGURE 4. Output signal timing tolerance.

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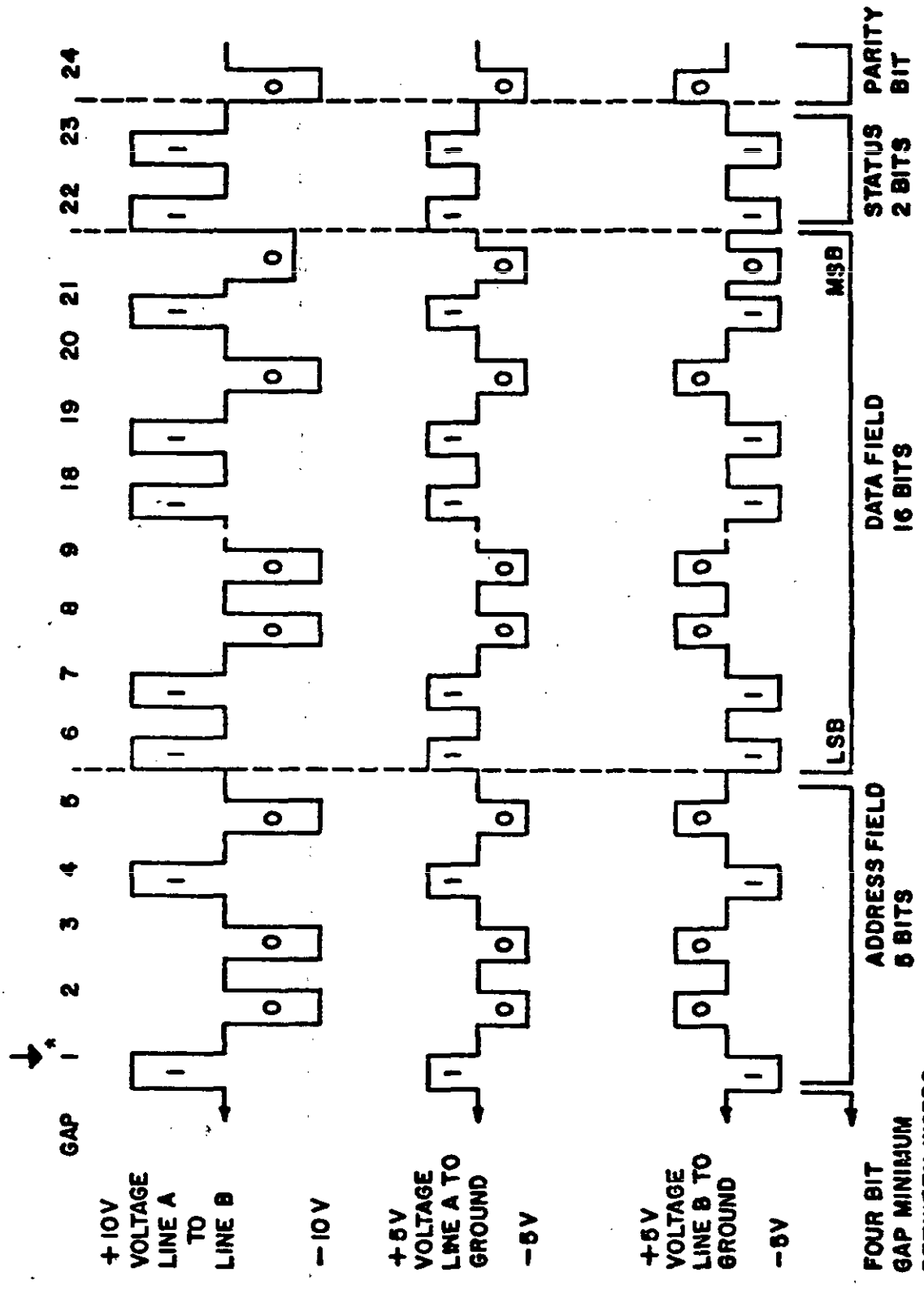


FIGURE 5. Digital voltage waveforms DRVS outputs.

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TABLE I. DRVS/CDU digital signal interface - test mode.

Parameter	Resolution	Range: Units	Address 1 2 3 4 5	Data 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21	Status 22 23	Parity 24
Beam velocity #1 - carrier (test)	7.8141 Hz/Bit	+64,005 Hz	1 0 1 1 0	LSB (18) (19) MSB SIGN →	M T	ODD
Beam velocity #2 - carrier (test)			1 1 1 1 0		M T	
Beam velocity #3 - carrier (test)			0 1 1 1 0		M T	
Beam velocity #4 - carrier (test)	↓	↓	0 0 1 1 0	↓ ↓ ↓	M T	
A/D test #1	4.8828X10^{-4}	-1.0000;+0.9995	0 0 0 1 1	(16) (17) MSB SIGN →	F T	
A/D test #2	4.8828X10^{-4}	-1.0000;+0.9995	0 1 0 1 1	(16 (17) MSB SIGN →	F T	
Cal constants K_Z, K_{YX} (test)	1/Bit	+127; -128	0 0 1 1 1	(12) (13) MSB SIGN →	F T	
Cal constants K_{XY}, K_{XZ}(test)	1/Bit	+127; -128	1 0 1 1 1	(12 (13) MSB SIGN) →	F T	
K_{SIN} (TAS)	4.8828X10^{-4}	-1.000;+0.9995	1 0 1 0 1	(16) (17) MSB SIGN →	S T	
K_{COS} (TAS)			0 0 1 0 1		S T	
K_{SIN} (heading)			1 1 1 0 1		F T	
K_{COS} (heading)			0 1 1 0 1		F T	
K_{SIN} (pitch)			1 1 0 0 1		F T	
K_{COS} (pitch)			0 1 0 0 1		F T	
K_{SIN} (roll)			1 0 0 0 1		F T	
K_{COS} (roll)			0 0 0 0 1		F T	
Bias B	↓	↓	1 0 0 1 1	↓ ↓ ↓	F T	
Antenna temperature	1°C/Bit	+127°C; -128°C	1 1 0 1 1	(12) (13) MSB SIGN →	F T	

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TABLE II. DRVS/CDU digital signal interface - NAV mode.

Parameter	Resolution	Range: Units	Address 1 2 3 4 5	Data 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21	Status 22 23	Parity 24
Beam velocity #1 + carrier	7.8141 Hz/Bit	+64,065 Hz	1 0 1 1 0	LSB ... (18) (19) MSB SIGN →	M T	ODD
Beam velocity #2 + carrier			1 1 1 1 0		M T	
Beam velocity #3 + carrier			0 1 1 1 0		M T	
Beam velocity #4 + carrier	↓	↓	0 0 1 1 0	↓ ↓ ↓	M T	
Cal constant ΔK_X	9.25X10^{-3} Hz/KT/Bit	-1.184; +1.175 Hz/KT	0 0 0 1 1	(12) (13) MSB SIGN →	F T	
Cal constant ΔK_Y	1.8816X10^{-2} Hz/KT/Bit	-2.408; +2.388 Hz/KT	0 1 0 1 1	(12) (13) MSB SIGN →	F T	
Cal constants ΔK_Z, K_{YX}	1.6525X10^{-1} Hz/KT/Bit(ΔK_Z)	-6.842; +6.737 Hz/KT	0 0 1 1 1	(9) ΔK_Z + SIGN; $+K_{YX}$ + (13) SIGN →	F T	
	5.63X10^{-3}/ Bit (K_{YZ})	-0.845; +0.839				
Cal constants K_{XY}, K_{XZ}	1.55X10^{-3}/ Bit (K_{YZ})	-0.625; +0.623	1 0 1 1 1	(16) $+K_{XY}$ SIGN; $+K_{XZ}$ + (13) SIGN →	F T	
	1.58X10^{-5}/ Bit (K_{YZ})	-0.006; +0.0045				
K_{SIN} (TAS)	4.8828X10^{-4}	-1.000; +0.9995	1 0 1 0 1	(16) (17) MSB SIGN →	S T	
K_{COS} (TAS)			0 0 1 0 1		S T	
K_{SIN} (heading)			1 1 1 0 1		F T	
K_{COS} (heading)			0 1 1 0 1		F T	
K_{SIN} (pitch)			1 1 0 0 1		F T	
K_{COS} (pitch)			0 1 0 0 1		F T	
K_{SIN} (roll)			1 0 0 0 1		F T	
K_{COS} (roll)			0 0 0 0 1		F T	
Bias B	↓	↓	1 0 0 1 1	↓ ↓ ↓	F T	
Antenna temperature	1°C/Bit	+127°C; -128°C	1 1 0 1 1	(12) (13) MSB SIGN →	F T	

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3.18.1.6 Range and bearing (see 4.11.6 and 4.11.15.2). The CDU shall compute and display range and bearing from aircraft present position to any one of 10 selected checkpoints when range is less than or equal to 999.9 km. The display resolution of range shall be ±0.1 km and that of bearing shall be ±1 degree.

3.18.1.7 Time en route (see 4.11.7). The CDU shall compute and display the estimate time en route from aircraft present position to any one of 10 selected checkpoints. This continuous computation shall be based on range to selected checkpoint and aircraft velocity. The display resolution of time shall be ±0.1 minute when time en route is less than 99.9 minutes.

3.18.1.8 Magnetic variation (see 4.11.8). The CDU shall be capable of entry, display and storage of magnetic compass variation associated with any one of 10 selected checkpoints and shall be used in the computation of present position. The display resolution shall be 0.1 degree. When magnetic variation is identical for each destination, only the first destination needs to have magnetic variation inserted.

3.18.1.9 Self test (see 4.11.9). The DNS shall incorporate two forms of self-test -- a continuous check of the CDU, and a test of the RTA, SDC when the TEST mode is selected. If a CDU failure occurs, the MAL light shall come on immediately, and when the MODE switch is set to TEST, a failure code shall be displayed. If no CDU failure has occurred and the TEST Mode is selected, the SDC shall transmit test data to the CDU for evaluation against stored data. If the system operation is satisfactory, a GO shall be displayed. If a failure is detected, the following diagnostic code shall be displayed:

Code	Failure indication
C (Followed by module failure numeric code)	CDU
R (Followed by module failure numeric code)	RTA
S (Followed by module failure numeric code)	SDC

3.18.1.10 BACKUP mode (see 4.11.10). When there is a loss of doppler velocity data, the CDU shall automatically enter the BACKUP mode and the MEM light shall illuminate. The CDU shall also provide a control for manual selection of the BACKUP mode.

3.18.1.11 Navigation Bench Test (see 4.11.15). The CDU shall be tested for navigation computations comprising of present position (LAT/LONG and UTM), velocities (V_X, V_Y, V_Z, V_H, V_V, V_D), and steering information. These tests shall be performed to assure that the ground track, ground speed, range, bearing, and present position coordinates display functions are functioning correctly and that the bench accuracy requirements are met.

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3.18.2 Controls, switch lamps and alphanumeric displays (see 4.11.11). The CDU controls and display shall be contained in a single unit that shall provide for the selection of operating modes and navigation displays. Any combination of CDU selector switch positions at any time shall not cause modifications to permanent memory or harm equipment associated with its use.

3.18.2.1 Control selectors (see 4.11.11). The following control selection shall be provided for entering, changing and enabling CDU functional modes.

3.18.2.1.1 Mode selector (see 4.11.11.1).

	Position	Function
a.	OFF	System is shut off.
b.	LAMP TEST	Checks operation of all CDU lamps.
c.	TEST	Initiates built-in-test exercise for the navigation set.
d.	UTM	Selects the Universal Transverse Mercator Reference System for Display and Entry.
e.	LAT/LONG	Selects the Latitude/Longitude Map Reference System for Display and Entry.
f.	BACKUP	Places the CDU in a Backup Mode of Operation using TAS plus remembered wind (if TAS is available) or remembered velocity (if TAS is not available). Permits pilot entry of best estimate of wind speed and direction in TAS operation. Also permits pilot entry of best estimate of ground speed and track angle for navigation computations. In this mode, data is displayed only in UTM coordinates.

3.18.2.1.2 Display selector (see 4.11.11.2).

	Position	Function
a.	WIND Sp/Dir	Selects wind speed and direction for display - wind speed in km/hr and wind direction in degrees. When the Mode Switch is set to BACKUP, wind speed and direction may also be entered manually.

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b.	XTK/TKE	Selects cross-track distance error (XTK) and track angle error (TKE) for display. XTK is displayed as right or left (R or L) in kilometers and tenths of a kilometer, and TKE is displayed as R or L in degrees. The R or L shall designate the direction of flight to reduce the error.
c.	GS/TK	Selects ground speed in km/hr and track angle (True) in degrees for display. When the MODE switch is set to BACKUP, ground speed and track angle may also be entered.
d.	PP	Selects present position coordinates for display. UTM or Lat/Long format is determined by setting of the MODE switch. The PP position is also used for present position updating on known landmarks.
e.	DIST/BRG/TIME	Enables display of distance, bearing (Magnetic) and time to the FLY-TO-DESTINATION. Distance is in kilometers and tenths of a kilometer; bearing is in degrees; and time is in minutes and tenths of a minute.
f.	DEST/TGT	Enables entry and display of DEST DISP/Thumbwheel destinations. UTM or LAT/LONG format is determined by the MODE switch.
g.	SPH/VAR	Enables entry and display of spheroid code and magnetic variation of destination selected on the DEST DISP thumbwheel.

3.18.2.1.3 Destination display selector thumbwheel (DEST DISP) (see 4.11.11.3).

Position	Function
Ø through 9, H, P	Used in conjunction with the DISPLAY SELECTOR for display of destination and present position coordinates, and spheroid and magnetic variations of selected destinations and present position.

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3.18.2.1.4 Fly-to-destination selector thumbwheel (FLY-TO-DEST) (see 4.11.11.4).

Position	Function
Ø through 9, H	Selects destination for which steering information is being computed.

3.18.2.2 Data entry keyboard (see 4.11.11.5). The following keyboard switch lamp controls shall be provided for data entry and enabling CDU navigation functions:

3.18.2.2.1 Switch lamp.

Switch Lamp	Function
a. Keyboard, A-Z and Ø-9	Provide the following data entry for CDU computer 1. Present position and destination coordinates and update 2. UTM spheroid 3. Magnetic variation 4. Wind speed and direction 5. Ground speed and track angle
b. KYBD (Keyboard)	Used in conjunction with the keyboard for data entry
c. CLR (Clear)	Clear data entry- pressed once clears last character entry; pressed twice, clears entire data panel under control
d. ENT (Enter)	Enter displayed data into CDU computer memory when in Keyboard mode.
e. TGT STR (Target Store)	Store present position coordinates in storage location displayed in the TGT STR indicator

3.18.2.3 Alphanumeric display (see 4.11.11.6). The CDU shall provide an alphanumeric display of navigational data. In addition to the following UTM or Latitude-Longitude position data, the display shall provide degree signs and decimal points.

3.18.2.3.1 UTM (see 4.11.11.6.1). The alphanumeric UTM readout shall be an instantaneous and continuous display consisting of the grid zone identifier (two numerics and an alpha), the 100 kilometer square identifier (2 alpha characters), and an eight-digit UTM metric readout.

3.18.2.3.2 Latitude/longitude (see 4.11.11.6.2). Position data

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in Latitude (N/S) and Longitude (E/W) shall be in degrees and minutes of arc with a resolution of 0.1 arc minutes.

3.18.2.3.3 DIM control (see 4.11.11.7). The alphanumeric display light intensity DIM control shall be located on the CDU front panel. Dimming for all other CDU front panel lettering, switch and status lamps shall be externally controlled (aircraft instrument panel dim control).

3.18.2.4 Status lamps (see 4.11.11.8).

3.18.2.4.1 Doppler memory (MEM) lamp. The MEM indicator shall illuminate when there is a loss of Doppler velocity data, thus indicating a memory condition (3.18.1.10).

3.18.2.4.2 MAL lamp. The MAL indicator shall illuminate when there is an equipment malfunction (3.18.1.9).

3.19 CDU digital inputs (see 4.11.12). The CDU input interface shall be compatible with the digital outputs described in 3.17.3 inclusive.

3.19.1 Circuitry. The CDU receiver circuit shall be of the balanced type as shown in figure 3. The input circuits shall be compatible with the SDC output circuit electrical characteristics described in 3.17.3.1 through 3.17.3.1.4 and shall have the following characteristics.

Resistance	-	12,000 ohms minimum
Capacitance	-	100 picofarads maximum

3.20 CDU digital output (see 4.11.13.1). The CDU shall provide the following digital outputs.

3.20.1 On/off discrete (see 4.11.13.1). The CDU ON/OFF discrete output shall be compatible with the ON/OFF discrete functions and input circuit requirements described in 3.17.1.2.

3.20.1.1 Circuitry (see 4.11.13.2). The ON/OFF discrete transmitting circuit shall be capable of accommodating any combination of the following load parameters:

Resistance	-	6,000 to 12,000 ohms
Capacitance	-	1,000 to 30,000 picofarads

3.20.1.2 Noise immunity (see 4.11.13.4). The CDU ON/OFF transmitting circuit shall be immune to noise pulses having a time-voltage integral of 1 volt amplitude in any period of 50 microseconds. These noise pulses are in addition to the signal levels set forth in 3.17.1.2.

3.20.1.3 Reverse current (see 4.11.13.3). The driver circuit shall be capable of withstanding up to one milliampere of reverse

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current without damage to the line driver components or circuitry.

3.20.2 CDU auxiliary digital output (AUX) (see 4.11.14). The AUX digital data output will consist of a two-wire circuit (twisted and shielded pair).

3.20.2.1 Electrical characteristics. The data word transmission shall use a bi-polar DC signal with a return to zero logic.

3.20.2.2 Voltage level (see 4.11.14.1). All digital voltage levels shall conform to the voltage levels of figure 2.

3.20.2.3 Circuitry (see 4.11.14.1). The AUX driver circuit shall be of the balanced type shown in figure 3. The output circuit shall be capable of accommodating any combination of the following load parameters.

Resistance	-	600 to 12,000 ohms
Capacitance	-	1,000 to 30,000 picofarads

3.20.2.3.1 Noise immunity (see 4.11.14.1.1). The AUX digital transmitter shall be immune to noise pulses having a time-voltage integral of 1 volt amplitude in any period of 50 microseconds. These noise pulses are in addition to the signal levels set forth in figure 2.

3.20.2.3.2 Reverse current (see 4.11.14.1.2). The driving current shall be capable of withstanding up to one milliampere of reverse current without damage to the line driver components or circuitry.

3.20.2.3.3 Digital data waveform (see 4.11.14.2). The digital data waveform shall be contained in a self synchronizing waveform as shown in figure 6. The waveform shall exhibit a voltage level transition at the beginning of each cycle or bit and return to zero in the middle of each cycle. The middle of the cycle shall be defined as being within ±10% of one half cycle as shown in figure 4.

3.20.2.4 Word structure (see 4.11.14.3). The standard word structure used for the AUX serial data bias shall consist of 32 bits. The AUX data word structure and its relation to the "CLOCK", "SYNC" and timing tolerances shall be as shown in figures 4, 6, and 7.

3.20.2.5 Standard language (see 4.11.14.4). A single transmitter shall be used for BCD data and the binary data. The data shall be encoded in 2 standard languages (BCD; 8-4-2-1 and Binary; two's complement) with the first bit used in each word to signal which language is being provided. The No. 32 bit shall be used for binary word parity check (figure 6).

3.20.2.5.1 Binary coded decimal (BCD) and binary output (see

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4.11.14.5). The BCD and Binary output shall be in accordance with table III and figure 8 (for Binary) and table IV and figures 7 and 8 (for BCD).

3.20.2.5.2 Code assignment for sign and status designation (see 4.11.14.6). A code matrix consisting of two bits from the standard word structure described in 3.20.2.4 shall be used to generate "TAGS" to identify "sign" associated with data and status of the equipment as shown in figure 8.

3.20.2.6 Synchronization. Word and bit synchronization shall be as follows:

3.20.2.6.1 Word synchronization (see 4.11.14.7). A blank consisting of the synch voltage on both AUX data transmission wires for a minimum period of four cycle shall exist between successive words that are transmitted on the same transmission system. This gap as shown in figure 6, shall provide the necessary synchronization for receiving equipment.

3.20.2.6.2 Bit synchronization (see 4.11.14.8). The clock or bit sync signal shall consist of a 11 ± 3.5 kHz square wave signal having a rise and fall time of 10 ± 5 microseconds as shown in figure 4.

3.21 Primary input power (see 4.12.1). The AN/ASN-128 ( ) shall meet all applicable requirements of MIL-STD-704, category B, and shall give specified performance from a 28 Vdc (100 watts maximum) power source as defined in MIL-STD-704 and shall meet the performance as specified herein for steady-state dc voltages for normal, abnormal and landing modes of electrical system operation as defined in table II, MIL-STD-704. No other power shall be required for operation except 26 Vac, 400 Hz synchro reference and a variable 5 Vac 400 Hz supply for edge lighting the Computer-Display Unit. In addition, the AN/ASN-128( ) shall give specified performance from a dc power source whose voltage drops to a minimum of 22.5 Vdc indefinitely. During periods when the primary power is outside of these limits, the system shall automatically shut down or otherwise protect itself against any damage or computer memory modifications. Short circuit protection for the power supply shall be provided.

3.21.1 Battery (see 4.12.2). Provisions shall be made for a battery to provide power to the non-volatile read-and-write memory (RAM) when the CDU MODE switch is in the OFF position. The CDU battery shall be case mounted in accordance with DL-SM-B-858970. The battery (BA-1090/U) shall be supplied as GFE (see 6.6). The mercury battery (BA-1090/U) shall be disposed of in accordance with Battery Disposition/Disposal Handbook, November 1988, Safety Office, USA CECOM, Ft. Monmouth, NJ.

3.21.2 Advisory lamp power (see 4.12.3). The CDU shall use a separate 28 Vdc advisory power source for illuminating advisory

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lamps. Steady-state advisory lamp power shall not exceed 8 watts.

3.21.3 Panel lamp power (see 4.12.4). The CDU shall use a separate 5 Vac 400 Hz panel power source for illuminating all edgelit panels and the keyboard (except the ENT key). This source can be continuously varied from 0 to 5 Vac in order to dim the panel lighting. The return for this source shall be through the front panel of the CDU per MIL-C-6781. The AN/ASN-128 panel lamp power requirements shall not exceed 8 watts.

3.21.4 Protection, electrical. Electrical overload protection shall be provided in compliance with requirement 8 of MIL-STD-454.

3.22 Burn-in/debugging (see 4.6.1). Each unit on order shall be subjected to a 48 hour (on time) burn-in with the last 24 (on time) hours failure free and meet the requirements herein, after subjection to the burn-in/debugging as specified.

3.23 Interchangeability (see 4.14). Like units, assemblies, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454 and shall be physically and functionally interchangeable. Items which are covered by specified electrical performance requirements shall meet such specified performance requirements prior to installation into the unit(s). After installation of these items and prior to submission of unit(s) for the quality performance specifications, adjustments shall be made if required, to permit the unit(s) to meet equipment specification requirements. Alterations of any items, or the unit(s) shall not be permitted.

3.24 Workmanship (see 4.13). The equipment shall be manufactured and assembled in accordance with the equipment drawings (see 3.2) and the following requirements and paragraphs herein:

MIL-STD-454	
Soldering	Requirement 5
Workmanship	Requirement 9

Paragraphs herein		
Cleaning	-	3.4
Controls	-	3.6
Securing of parts	-	3.8
Soldering	-	3.10
Wiring, cabling, bonds and grounds	-	3.11

3.25 System safety engineering (see 4.13).

3.25.1 Safety hazards. Safety hazards shall be kept to a minimum in accordance with MIL-STD-454, requirement 1 and the safety hazard requirements of MIL-STD-1472.

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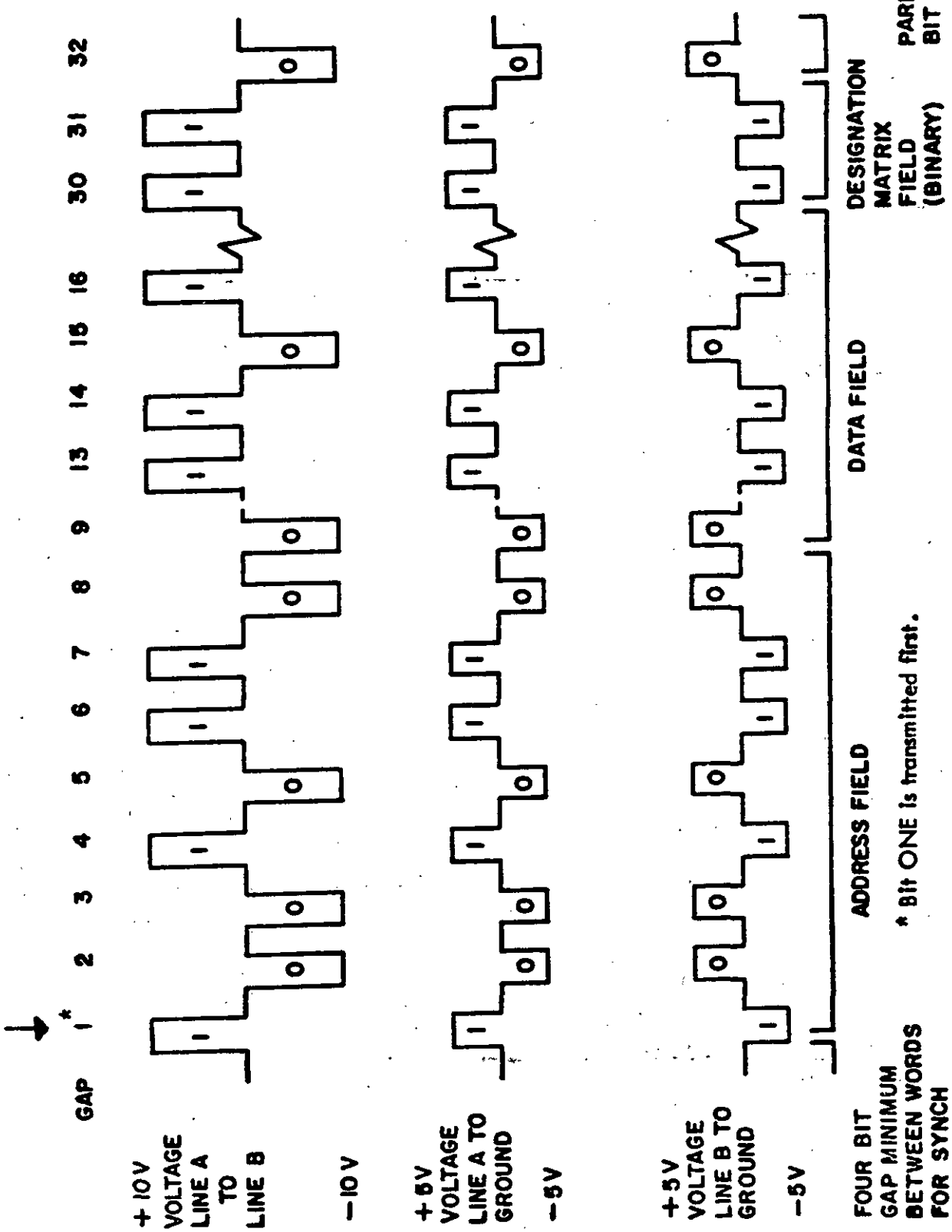


FIGURE 6. Digital voltage waveforms DNC auxiliary outputs.

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FIGURE 7. BCD Word Structure.

Parameter	Reso-lution	Range	Example	Sig fig	Pad fig	32-Bit BCD word Structure MSB ← Information → LSB ← Label →
						32 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1
Distance to go	0.01	+999.99	981.73	5	1	0 0 0 0 1 0 0 1 1 0 0 0 0 0 0 1 0 1 1 1 0 0 1 1 0 1 1 0 1 0 0 0
Cross Track deviation	0.1	±399.9	118.6	4	2	0 0 0 1 0 0 0 1 1 0 0 0 0 1 1 0 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0 0
Desired track	0.1	0-359.9	258.3	4	2	0 0 1 0 0 1 0 1 1 0 0 0 0 0 0 1 1 0 0 0 0 0 0 0 0 1 1 0 0 0 0 0
Track angle error	0.1	0 - ±180.0	-095.7	4	2	1 1 0 0 1 0 0 1 0 1 0 1 0 1 1 1 0 0 0 0 0 0 0 0 1 0 1 0 0 0 0 0
Ground speed	1.0	0 - 999	650	3	3	0 0 0 0 0 1 1 0 0 1 0 1 0 0 0 0 0 0 0 0 0 0 0 0 0 1 0 1 0 0 0 0
Position (UTM) northing	0.01	0 - 99.99	87.32	4	2	0 0 0 0 1 0 0 0 0 1 1 1 0 0 1 1 0 0 1 0 0 0 0 0 0 1 0 1 1 0 0 0
Position (UTM) easting	0.01	0 - 99.99	96.89	4	2	0 0 0 0 0 0 0 0 0 1 1 0 1 0 0 0 0 0 0 1 0 0 0 0 0 1 0 0 1 1 0 0
Heading velocity (V_H)	0.1	0 - ±999.9	+063.2	4	2	0 0 0 0 1 0 0 1 0 1 1 0 0 0 1 1 0 0 1 0 0 0 0 0 1 0 1 0 1 1 1 0
Drift velocity (V_D)	0.1	0 - ±999.99	-016.8	4	2	1 1 0 0 0 0 0 0 0 0 0 1 0 1 1 0 1 0 0 0 0 0 0 0 0 1 1 0 1 1 1 0
Vertical velocity (V_V)	0.1	0 - ±999.9	-063.7	4	2	0 0 0 0 0 0 0 0 0 0 0 0 0 0 1 1 0 1 1 1 0 0 0 0 1 1 1 0 1 1 1 0
100 km square id	---	A - V	FT	4	2	0 0 1 0 0 0 1 1 0 1 0 1 0 1 0 0 0 0 0 0 0 0 0 0 0 0 1 0 0 0
(Alphabetic letters)	---	(Except I & O)				F T

Note:

1. 0 - PAD BIT ∧ = Decimal Point Position
2. Bit 1 is transmitted first.

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BCD designation matrix (no parity provided) bit no.		(BNR) binary designation matrix		Designation
32	31	31	30	
0	0	0	0	Plus (+) north, east, right, to
0	1	0	1	Failure warning
1	0	1	0	No computed data
1	1	1	1	Minus (-) south, west, left, from

NOTES:
1. Both bits should be "zero" when no sign is needed, providing data is valid.
2. Bit no. 32 is used for parity check in BNR.
3. Failure Warning indicates that a failure has occurred and the digital output word is not valid.
4. No Computed Data indicates computed data is not available for any reason, but is distinguished from a failure warning by the fact that under certain circumstances, computed data may not be available even though no failure has been detected within the equipment.

FIGURE 8. Sign and status designation.

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TABLE III. CDU auxiliary digital output signals: Binary output.

	Minimum update rate	Pad bits 1/	Data Bits	Units	Range	Address 8 7 6 5 4 3 2 1
True heading - Sin	7.32 /sec	11	10	N/A	±1.0	0 0 0 1 1 1 1 1
Cosine	7.32 /sec	11	10	N/A	±1.0	1 0 0 1 1 1 1 1
Roll - sine	7.32 /sec	11	10	N/A	±1.0	0 1 0 1 1 1 1 1
Cosine	7.32 /sec	11	10	N/A	±1.0	1 1 0 1 1 1 1 1
Pitch - sine	7.32 /sec	11	10	N/A	±1.0	0 0 1 1 1 1 1 1
Cosine	7.32 /sec	11	10	N/A	±1.0	1 0 1 1 1 1 1 1
Heading velocity (V_H)	7.32 /sec	6	15	km/hr	±3276.7	1 0 1 0 1 1 1 1
Drift velocity (V_D)	7.32 /sec	6	15	km/hr	±3276.7	0 1 1 0 1 1 1 1
Vertical velocity (V_V)	7.32 /sec	6	15	km/hr	±3276.7	1 1 1 0 1 1 1 1
Velocity along longitudinal (V_X) axis	7.32 /sec	6	15	km/hr	±3276.7	0 1 0 0 1 1 1 1
Velocity along lateral axis (V_Y)	7.32 /sec	6	15	km/hr	±3276.7	1 1 0 0 1 1 1 1
Velocity along yaw axis (V_Z)	7.32 /sec	6	15	km/hr	±3276.7	0 0 1 0 1 1 1 1
Bearing to destination 2/	3.66 /sec	10	11	deg	0±180°	1 1 1 0 1 0 0 1
Vertical pointer - SHIU	3.66 /sec	6	15	μA	0±225	0 0 0 0 1 1 1 1
Side pointer - SHIU	3.66 /sec	16	15	μA	0±225	1 0 0 0 1 1 1 1

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TABLE III. CDU auxiliary digital output signals: Binary output- Continued.

	Minimum update rate	Pad bits 1/	Data Bits	Units	Range	Address 8 7 6 5 4 3 2 1
Horizontal pointer SHIU	3.66 /sec	6	15	μA	0±225	1 1 1 1 0 1 1 1
Bite word	3.66 /sec	-	-	-	-	1 1 0 1 0 1 1 1
Course dev. (XTK) (DIST)	3.66 /sec	11	10	nA	±1.0	1 0 1 1 0 1 1 1
BHD (bearing to destination from aircraft heading)	3.66 /sec	10	11	deg	±180	0 1 1 1 0 1 1 1

1/ Pad bits are added to the least significant end of the data. These may be "0" bits, characters or valid data.

2/ Bearing to destination shall be computed with magnetic north as reference.

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TABLE IV. CDU auxiliary digital output signals: serial BCD output.

	Minimum update rate	Sig fig	Pad fig	Units	Resolution	Range	Address 8 7 6 5 4 3 2 1
Distance to go	3.66 /sec	5	1	km	0.01	0 - 999.99	0 1 1 0 1 0 0 0
Cross track Deviation	3.66 /sec	4	2	km	0.1	±399.9	1 1 0 0 0 0 0 0
Desired track	3.66 /sec	4	2	deg	0.1	0-359.9	0 0 1 0 0 0 0 0
Track angle error	3.66 /sec	4	2	deg	0.1	0-±180	1 0 1 0 0 0 0 0
Ground Speed	3.66 /sec	3	3	km/hr	1.0	0 - 999	0 1 0 1 0 0 0 0
Position (UTM) northing	3.66 /sec	4	2	km	0.01	0-99.99	0 1 0 1 1 0 0 0
Position (UTM) easting	3.66 /sec	4	2	km	0.01	0-99.99	1 0 0 1 1 0 0 0
Drift velocity (V)	3.66 /sec	4	2	km/hr	0.1	±999.9	0 1 1 0 1 1 1 0
Vertical Velocity (V)	3.66 /sec	4	2	km/hr	0.1	±999.9	1 1 1 0 1 1 1 0
100 km sq id 1/ (Alphabetic letters)	3.66 /sec	2	1	--	--	A - V accept I an O	0 0 0 1 1 0 0 0
Heading Velocity (V)	3.66 /sec	4	2	km/hr	0.1	±999.9	1 0 1 0 1 1 1 0
Bite word	3.66 /sec	-	-	-	-	-	0 0 1 0 1 0 0 0

1/ 7 BIT ASCII

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3.25.2 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, lenses).

3.25.3 Finished items. Finished items shall bear no raw, sharp or rough edges on any parts.

3.26 Lamps (see 4.17). Display lamps shall have a luminance requirement of 50 ± 15 millicandelas for the numeric and 90 ± 25 millicandelas for the alpha with all segments energized at rated voltages, and shall meet the requirements of Particle Impact Noise Detection Test Method of 3.4 in MIL-STD-883.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of this specification. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (does not include packaging). (see 4.3)

b. Inspection covered by subsidiary documents. (see 4.4)

c. Quality conformance inspection of equipment before packaging (see 4.5).

4.3 First article (see 3.3 and 6.2). Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor.

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4.3.1 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, burn-in, and the inspections specified for group A, group B, group C and group D (see tables VI, VII, VIII and IX, respectively). The inspection shall be performed in the order indicated in table V. Other first article tests may precede, follow or be interspersed among the foregoing.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the appropriate subsidiary documents, as part of the inspection before preparation for delivery:

Inspection	Requirement paragraph
Protective finish	3.7
Marking	3.9
Parts, materials and processes	3.12

4.5 Quality conformance inspection of equipment before packaging. Each unit shall be subjected to burn-in/debugging (see 3.22). The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement after burn-in/debugging shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government at its discretion, may perform all or any part of the specified inspections to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor at the contractor's facility.

4.5.1 Group A inspection. Each unit on contract shall be inspected for conformance to the inspections specified in table VI. Factors of lot composition not defined herein, or in the contract, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105 using the general inspection levels indicated in table I of MIL-STD-105. Group A visual and mechanical inspection of modules and subassemblies may be accomplished in process. Following burn-in, a visual and mechanical inspection of the completed assembly shall be performed, including removal of covers for internal inspection. It is not necessary to disassemble the unit for reinspection, beyond removal of covers, to satisfy the visual and mechanical inspection requirements provided modules and subassemblies have been inspected in process.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

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4.5.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. This inspection shall conform to table VII and to the special inspection levels of table I of MIL-STD-105.

4.5.2.1 Group B sampling plan. The group B AQL shall be 6.5% and the inspection level shall be S4, except for the velocity and navigation accuracy test paragraphs 4.10.7 through 4.10.7.1.5 which shall be performed on at least two systems every 200 systems or every six months whichever comes first.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

4.5.3 Group C inspection. This inspection shall be as listed in table VIII and shall be performed on sample unit(s) that have been subjected to and met group A and group B inspections. No defects shall be allowed. Sample unit(s) shall be selected in accordance with 4.5.3.2.

4.5.3.1 Order of inspection within Group C. Group C inspections shall be performed in the order shown in each subgroup in table VIII.

4.5.3.2 Sampling for Group C inspection. Units shall be selected at random for each group of inspections listed in table VIII as specified in 4.5.3.2.1 through 4.5.3.2.4. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot(s).

4.5.3.2.1 Subgroup I. For this subgroup, two units from the first forty (40) production units after first article inspection shall be selected for the first group C inspection. For subsequent inspection, two units from each group of a minimum of 120 units shall be selected.

4.5.3.2.2 Subgroup II. For this subgroup, two units (these units may be the units selected for above subgroup I), from the first forty (40) production units after first article inspection shall be selected for the first group C inspection. For subsequent group C inspection, two units from each group of 120 units shall be selected, except that temperature-altitude shall be performed on two units from each group of a minimum of 120 units.

4.5.3.2.3 Subgroup III. For this subgroup, two units from the first forty (40) production units after first article inspection, shall be selected for the first group C inspection. For subsequent group C inspection, two units from each group of 120 units shall be

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selected, except that fungus, acceleration, explosion, dust, and rain shall be performed on two units from each group of a minimum of 210 units.

4.5.3.2.4 Subgroup IV. For this subgroup, one (1) unit from the first (50) production units after first article inspection shall be selected and tested for compliance with the requirements of paragraphs 3.11.8, 3.11.9 and 3.15.1. For subsequent group C inspections, one (1) unit shall be selected from each group of 50 units starting with the 51st and tested for compliance with the requirements of paragraph 3.11.8, 3.11.9 and CE04, RE02, and RE02.1 of paragraph 3.15.1 only.

4.5.3.2.5 Subgroup V. For this group two each RTA units selected at random from each production lot of 50 shall be tested per 4.8.12.

4.5.3.3 Non-Compliance. Actions required relative to group failures shall be as specified in the contract or purchase order.

4.5.3.4 Re-inspection of conforming group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group C inspection may be delivered on the contract, provided all damage is repaired, cleaned of all foreign matter, and the sample units are resubjected to and pass group A inspection.

4.5.4 Group D inspection. This inspection shall consist of the test specified in table IX and shall be performed on units from lots which have successfully completed the burn-in/debugging and have been subjected to and met group A and group B inspections.

4.5.4.1 Sampling for group D inspection. For group D inspection, three to six samples shall be randomly selected by an authorized Government representative for the production reliability tests (4.6.2) from units in the lot of 100 units after first article and every 100 units thereafter.

4.5.4.2 Group D failures. Actions required relative to group D failures shall be as specified in the contract or purchase order.

4.5.4.2.1 Re-inspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and passed group D inspection may be delivered on the contract, provided all damage is repaired, cleaned of all foreign matter, and the sample units are resubjected to and pass group A inspection.

4.6 Reliability (see 3.14).

4.6.1 Burn-in/debugging (see 3.22). Each unit on order shall be subjected to a 48 hour (on time) burn-in period. The last 24 hours (on time) of burn-in/debugging shall be failure-free. The

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burn-in/debugging shall be conducted under the conditions of paragraph 4.6.2.2 and all unit(s) shall be exercised and monitored in the same manner as during reliability testing. The failures which occur during the burn-in shall be recorded but shall not be considered relevant for computation of equipment MTBF. A failure shall be as defined in paragraph 4.6.2.3.

4.6.2 Reliability testing. (Group D inspection) The contractor shall conduct all reliability tests in accordance with test plan XXI of MIL-STD-781. The minimum test time per equipment shall be in accordance with MIL-STD-781.

4.6.2.1 Conditions precedent to reliability testing. Prior to reliability testing, the contractor shall perform the burn-in of 4.6.1 and the thermal and vibration surveys in accordance with MIL-STD-781.

4.6.2.2 Reliability test conditions. All reliability testing and burn-in shall be conducted under the conditions of test level E of MIL-STD-781 as follows:

a. The temperature shall be cycled from -54° to +55°C in accordance with MIL-STD-781.

b. The equipment shall be vibrated at 2.2g ± 10% at any non-resonant frequency between 20 Hz and 60 Hz, measured at the equipment case. The duration of vibration shall be at least 10 minutes of each operating hour.

c. The input voltage cycling shall be accomplished as follows: The input voltage shall be maintained at 30.25 Vdc for 1 complete temperature cycle, at 27.5 Vdc for 1 complete temperature cycle and at 24.75 Vdc for 1 complete temperature cycle. This cycling shall be repeated continuously throughout the test.

d. The unit(s) shall operate in the navigate mode. The equipment shall be on-off cycled once during each complete temperature cycle.

4.6.2.3 Failure definition. A failure is defined as the inability of the equipment to perform its required function. This includes any event which causes departure from the performance as required by this specification. MIL-STD-781 shall apply. Lamp and bulb failures shall not be considered relevant failures.

4.6.2.4 Parameters to be measured. As a minimum the following parameters shall be measured and recorded twice daily: once during high temperature stabilization, once during low temperature stabilization. Vibration shall not be avoided during measurement.

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Parameter	Requirement Paragraph	Test Paragraph
XTK, TKE	3.18.2.1.2	4.11.15.2
Navigation bench test	3.18.1.11	4.11.15
Ground track-ground speed	3.18.1.5	4.11.15.1
Range and bearing	3.18.1.6	4.11.15.2
Self test	3.18.1.9	4.11.9.1

4.6.2.5 Rework and retest provision. Once a reject decision has been reached, no rerun can be made of that test on the same or a new sample from the same lot, unless positive action, consisting of design change, modification, or rework actually relating to observed failures has been incorporated. Any subsequent return must start at zero hours operating time.

4.7 Standard test conditions. Unless otherwise specified, inspections shall be performed under the following conditions:

a. Input power: 28.0 ± 2.0 Vdc

b. Warm-up Time: Not greater than 5 minutes

c. Relative humidity: Room ambient

d. Temperature: Room ambient

e. Atmospheric pressure: Room ambient

4.8 Service conditions (see 3.13). Unless otherwise specified, the service condition tests shall be conducted in accordance with MIL-STD-810. Electrical tests and inspections shall be conducted as specified in table X. No defects shall be allowed.

4.8.1 Temperature-altitude. The unit(s) shall be subjected to the operating and the non-operating, altitude and temperature range with supply voltages from 22.5 through 30 Vdc as specified in test method 520.0 of MIL-STD-810, except that the temperature in steps 5 and 7 shall be 85°C and 71°C, respectively. In addition, a one-hour exposure (non-operating) to a pressure of 5.54 inches of Hg, simulating air transport at 40,000 feet and 23°C ± 10°C shall be included. These tests shall be made as indicated to determine compliance with 3.13.1 for class 1A equipment.

4.8.2 Humidity. The unit(s) shall be subjected to the test of method 507, procedure II. In step 6 of procedure II, measurements shall be taken during the last 5 hours of last test cycle. These tests shall be made as indicated to determine compliance with 3.13.2.

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TABLE V. First article tests.

Rqmt para	Test para	RTA, SDC, and CDU sample size				
		lea RT-1193 CV-3338 CP-1252	lea RT-1193 CV-3338 CP-1252	lea RT-1193 CV-3338 CP-1252	lea RT-1193 CV-3338 CP-1252	RT-1193 CV-3338 CP-1252[a]
3.7, 3.9 and 3.12 subsidiary documents	4.4	1	1	1	1	1
3.5 Weight	4.13	2				
3.22 Burn-in	4.6.1	3	2	2	2	2
4.5.1 Group A inspection	(See table VI)	4	3	3	3	3
4.5.2 Group B inspection	(See table VII)		4			
3.13.1 Temperature - altitude	4.8.1	5				
3.13.5.1 Bench handling	4.8.5.1		5			
3.13.5.2 Equipment	4.8.5.2		6			
3.13.5.3 Crash safety[b]	4.8.5.3					
3.13.4 Vibration	4.8.4		7			
3.13.10 Rain	4.8.10	6				

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TABLE V. First article tests - Continued.

Rqmt para	Test para	RTA, SDC and CDU sample size lea RT-1193 CV-3338 CP-1252	lea RT-1193 CV-3338 CP-1252	lea RT-1193 CV-3338 CP-1252	lea RT-1193 CV-3338 CP-1252	RT-1193 CV-3338 CP-1252
3.13.2 Humidity	4.8.2		8			
3.15 EMI	4.8.11				4	
3.13.9 Acceleration	4.8.9			4		
3.13.8 Explosive atmosphere	4.8.8			5		
3.13.3 Dust	4.8.3		9			
3.13.7 Salt fog	4.8.7		10			
3.13.6 Fungus	4.8.6			6		
3.13.11 RTA seal	4.8.12	7			5	
3.14 Reliability (Group D inspection)	4.6.2 4.5.4					4

a. A maximum of 4 units shall be selected for first article reliability tests.

b. The crash safety test may be performed using a dummy load. (see 6.5)

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TABLE VI. Group A inspection.

Inspection	Reqt para	Insp para	AQL	
			Major	Minor
Visual and mechanical	3.24	4.13		
Modules and frames			0.65%	2.5%
RT-1193/ASN-128( )				
Reference signal generator				
Receiver/transmitter				
Wave guide switch				
Electronic module				
Antenna assembly				
CV-3338/ASN-128( )				
Leakage elimination filter				
Frequency tracker				
Timer/interface				
A/D converter				
Power supply				
CP-1252/ASN-128( )				
Center processing unit				
RAM/ROM				
I/O				
Numeric logic				
Display logic				
Post regulator				
Assemblies				
RTA, RT-1193/ASN-128( )			1.0%	4.0%
SDC, CV-3338/ASN-128( )			1.0%	4.0%
CDU, CP-1252/ASN-128( )			1.0%	4.0%
Electrical				
RTA, RT-1193/ASN-128( )				
RTA input/output interface	3.16.8	4.9.1.6.1	1.0% combined 1/	
SDC, CV-3338/ASN-128( )				
Primary power	3.21	4.12.1.1		

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TABLE VI. Group A inspection - Continued

Inspection	Reqt para	Insp para	AQL	
			Major	Minor
Velocity test (SDC)	3.17.3.6	4.11.1.2		
Altitude/attitude limits	3.16.6, 3.16.7	4.9.1.5.b 4.10.8	1.0% combined 1/	
SDC digital output	3.17.3	4.10.5.1		
CDU, CP-1252/ASN-128( )				
Area of operation	3.18.1.1	4.11.1.1		
Navigation Bench Test	3.18.1.11	4.11.15		
Controls, switch lamps and alphanumeric displays	3.18.2	4.11.1.a 4.11.3, 4.11.4, 4.11.11.1.1, 4.11.11.2.2, 4.11.11.2.3, 4.11.11.2.4,	1.0% combined 1/	
Test	3.18.1.9	4.11.9.1 through 4.11.9.4		
CDU digital output	3.20	4.11.13.1		
CDU aux digital data output	3.20.2	4.11.14.1.a and 4.11.14.2.a		
Immersion				
RTA, RT-1193/ASN-128	3.13.12	4.8.13		

1/ All electrical defects are considered major.

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TABLE VII. Group B inspection.

Inspection	Reqt para	Insp para	AQL
Electrical			
RT-1193/ASN-128( )			
Frequency	3.16.2	4.9.1.1	6.5% for the
Radiated power	3.16.3	4.9.1.3	combined group
Navigation accuracy test	3.16.5 3.16.5.1	4.10.7 through 4.10.7.6	
CV-3338/ASN-128( )			
Signal acquisition	3.17.2.2	4.10.4	6.5% for the
Analog signal inputs	3.17.1	4.10.1 through 4.10.3	combined group
SDC digital output	3.17.3	4.10.5	
CP-1252/ASN-128( )			
CDU digital output	3.20	4.11.13	
CDU auxillary digital outputs	3.20.2	4.11.14	6.5% for the combined group
Area of operation	3.18.1.1	4.11 through 4.11.11	
Mechanical interchangeability	3.23	4.14	6.5%
Electrical interchangeability	3.23	4.14	6.5%

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TABLE VIII. Group C inspection.

RT-1193/ASN-128( ) CV-3338/ASN-128( ) CP-1252/ASN-128( )		
Inspection	Para	Para
Subgroup I		
Input power	3.21	4.12 4.12.1.2
Subgroup II		
Temperature-altitude	3.13.1	4.8.1
Humidity	3.13.2	4.8.2
Vibration	3.13.4	4.8.4
Subgroup III		
Dust	3.13.3	4.8.3
Bench handling shock	3.13.5.1	4.8.5.1
Equipment shock	3.13.5.2	4.8.5.2
Crash safety shock	3.13.5.3	4.8.5.3
Fungus	3.13.6	4.8.6
Salt fog	3.13.7	4.8.7
Explosive atmosphere	3.13.8	4.8.8
Acceleration	3.13.9	4.8.9
Rain	3.13.10	4.8.10
Weight	3.5	4.15
Subgroup IV		
EMI	3.15	4.8.11
Subgroup V		
RTA seal	3.13.11	4.8.12

TABLE IX. GROUP D inspection.

Inspection	Test paragraph	Requirement paragraph
Reliability	4.6.2	3.14

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TABLE X. Environmental tests and inspection.

Environmental Test	Temp. alt.		Humid.		Dust		Vib.		Shock						Fungus		Salt fog		Exp atmos		Accel		Rain	
									Equip		Crash safety		Bench handl											
Equipment Reqt para / Test para	Before and after	During	Before and after	During	Before and after	During	Before and after	During	Before and after	During	Before and after	During	Before and after	During	Before and after	During	Before and after	During	Before and after	During	Before and after	During	Before and after	During
3.18.1.1 Navigation test / 4.11.1.1	X	NA	X	NA	X	NA	X	NA	X	NA	NA	NA	NA	NA	NA	NA	NA	NA	X	NA	X	NA	X	NA
3.17.3.6 Velocity test (SDC) / 4.11.1.2	X	NA	X	NA	X	NA	X	NA	X	NA	NA	NA	NA	NA	NA	NA	NA	NA	X	NA	X	NA	X	NA
3.16.2 Radiated Frequency / 4.9.1.1	X	NA	X	NA	X	NA	X	NA	X	NA	NA	NA	NA	NA	NA	NA	NA	NA	X	NA	X	NA	X	NA
3.16.1 Radiated power / 4.9.1.3	X	NA	X	NA	X	NA	X	NA	X	NA	NA	NA	NA	NA	NA	NA	NA	NA	X	NA	X	NA	X	NA
3.17.3 SDC digital data output / 4.10.5.1 & 4.10.6.a	X	NA	X	NA	X	NA	X	NA	X	NA	NA	NA	NA	NA	NA	NA	NA	NA	X	NA	X	NA	X	NA
3.18.2 Controls, switch lamps and alphanumeric displays / 4.11.1.a, 4.11.3, 4.11.4, 4.11.11.1.1, 4.11.11.2.2, 4.11.11.2.3, & 4.11.11.2.4	X	NA	X	NA	X	NA	X	NA	X	NA	NA	NA	NA	NA	NA	NA	NA	NA	X	NA	X	NA	X	NA
3.20.2 CDU auxiliary digital data output / 4.11.14.1a & 4.11.14.2a	X	NA	X	NA	X	NA	X	NA	X	NA	NA	NA	NA	NA	NA	NA	NA	NA	X	NA	X	NA	X	NA
3.18.1.9 Self test / 4.11.9.1	X	(X)	X	(X)	X	NA	X	X	X	NA	NA	NA	X	NA	NA	NA	NA	NA	X	NA	X	NA	X	NA
3.6, 3.8 Visual and mechanical inspection / 4.13	X	NA	X	NA	X	NA	X	NA	X	NA	NA	NA	NA	NA	NA	NA	NA	NA	X	NA	X	NA	X	NA

(X) Denotes intermittent test.

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4.8.3 Dust. The internally mounted units shall be subjected to the test of method 510, procedure I, with the exception that the air velocity in steps 1, 2, 3 shall be 200 ± 100 feet per minute. The externally mounted units shall be subjected to the test of method 510, procedure I. These tests shall be made as indicated to determine compliance with 3.13.3.

4.8.4 Vibration. The DNS unit(s) shall be subjected to the test of method 514.2, procedure I, part I, category C, curve M of figure 514.2-3 truncated at 500 Hz, with the unit normally mounted without the use of additional supporting pins or bracket. The time schedules for the tests shall be in accordance with table 514.2-III, except the sweep time shall be 30 minutes. These tests shall be made as indicated to determine compliance with 3.13.4.

4.8.5 Shock (see 3.13.5).

4.8.5.1 Bench Handling. The unit(s) shall be subjected to the test method 516.1, procedure V, to determine compliance with 3.13.5.1.

4.8.5.2 Equipment. The unit(s) normally mounted shall be subjected to the tests of method 516.1, procedure I, with shock pulses in accordance with figure 516.1-2 for flight vehicle equipment. These tests shall be made to determine compliance with 3.13.5.2.

4.8.5.3 Crash safety. The unit(s) normally mounted shall be subjected to the tests of method 516.1, procedure III, with shock pulses, in accordance with figure 516.1-2 for flight vehicle equipment. These tests shall be made, as indicated, to determine compliance with 3.13.5.3.

4.8.6 Fungus (see 3.13.6). The unit(s) shall be subjected to 28 days of fungus test as specified in method 508, procedure I of MIL-STD-810. All external covers shall be opened and the interiors sprayed with the specified mixed spore suspension. The unit(s) shall be subjected to the test in this open condition and as specified to determine compliance with 3.13.6. The unit(s) shall not be required to operate. The equipment test model shall not be specially cleaned for the fungus test, except for the cleaning it receives during or after production. If it is necessary for the manufacturer after assembly to remove accumulated production and handling contaminants by cleaning prior to packaging and delivery, the equipment shall not be tested for three days after cleaning, to allow the complete evaporation of the cleaning compound. After 14 and 28 days of test, the control items per 3.1.3 and 3.1.4 of method 508 of MIL-STD-810 shall show profuse growth over at least 50% of the area of the control items. After 28 days, based upon visual examination, the equipment assembly shall pass if it shows no more than sparse microbial growth with restricted tuberal growth in an area of 1 to 10% or less of the total area and no more than six

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unrelated minute colonies with mycelial development in areas only in other than critical circuit portions; such as terminal spacings, printed circuit board, etc. with sparse growth due to random contamination or traces of unmixed material ingredients. The equipment shall fail if it shows more than the growth specified above. The test shall be repeated if the control items fail to show profuse growth after 14 and 28 days of test.

4.8.7 Salt fog. The unit(s) shall be exposed in its normal operating condition to salt fog in accordance with method 509, procedure I, for 48 hours. The unit(s) shall not be opened. The unit(s) shall not be required to operate electrically. All external, movable controls, access covers and test buttons shall be operated mechanically and there shall be no corrosion of finishes and metals. Such corrosion shall be defined as any visible degradation of the test item surfaces that can be attributed to flaky, pitted blistered, or otherwise loosened finish or metal surface. These tests shall be made as indicated to determine compliance with 3.13.7.

4.8.8 Explosive atmosphere. The unit(s) shall be tested for explosion in accordance with method 511, procedure 1, to determine compliance with 3.13.8.

4.8.9 Acceleration. The unit(s) shall be tested in accordance with method 513.1, procedure II. The unit(s) shall be subjected, in turn, in each direction along each of its three orthogonal axis. The test level shall be 6g's for at least one minute. These test shall be made to determine compliance with 3.13.9.

4.8.10 Rain. The internally mounted units (SDC and CDU), with cables in place, shall be subjected to test method 506.1, procedure II, using distilled water for rain simulation. After completion of the rain test, the units shall be turned on within 30 minutes and shall be tested in accordance with table X. The externally mounted unit (RTA) shall be subjected to the test of method 506, procedure I. During each 30 minute rain period, the wind shall be maintained for 15 minutes, commencing 5 minutes after the start of the period as specified in the procedure. The unit(s) shall be operated the last 10 minutes of the last 30 minute period and shall meet the tests of table X in compliance with 3.13.10.

4.8.11 Electromagnetic interference. The unit(s) shall be tested utilizing the measurement techniques of MIL-STD-462, notice 3. All testing shall be performed with interconnecting cables of at least 30 foot in length between components of the DNS. These tests are performed to determine compliance with 3.15.

4.8.12 RTA seal. Two sealed RTA units shall be tested per the following steps:

a. Operational self-test per 3.18.1.9 and 4.11.9.

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b. Temperature/vibrate samples by running these cycles as defined in figure 1 of burn-in test procedure Y201A551Y010, and 4.6.2.2a and b.

c. Self-test step (a) above.

d. Dry for 24 hours at 77°F (25°C) at 50% RH.

e. Weigh to within three grams accuracy.

f. Immersion test per method 512, procedure 1, MIL-STD-810, except as follows:

1. Immersion media shall be ordinary tap water.

2. The RTA shall be completely immersed, just above the antenna cover, but below the connector interface (with a sealed cap on the connector).

g. Wipe and repeat step d above.

h. Weigh per step e above and calculate weight gain, if any.

i. Self-test per step a above.

4.8.13 RTA immersion (see 3.13.12). Each RTA on contact shall be tested per the following:

a. Inject regulated dry air into the RTA until the RTA is pressurized to 0.75 ± 0.1 psi of positive pressure.

b. Immerse the RTA into a container of water until the entire unit, excepting the output connector, is covered with water.

c. Remove air bubbles that may be adhering to the RTA.

d. Maintain 0.75 ± 0.1 psi positive pressure while the RTA is immersed. The RTA shall remain immersed for a test period of three minutes.

e. Observe the RTA for the formation of any bubbles from any one location during the test period shall be cause for rejection of the unit under test.

4.9 Receiver-Transmitter-Antenna, Radar, RT-1193/ASN-128( ) (see 3.16).

4.9.1 RTA beams; beam switching (see 3.16.1.1). With the RTA operating normally, the radiated antenna output shall be detected to

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verify that the beams are being switched in the proper sequence (figure 1). The beam switching rate shall also be measured and shall not exceed 7.5 Hz.

4.9.1.1 Radiated frequency (see 3.16.2). Measure the frequency of the radiated power. The frequency shall be between 13.250 GHz and 13.400 GHz.

4.9.1.2 Modulation index (see 3.16.2.1). Measure the transmitter modulation index. The index shall be 0.92 ± 15%.

4.9.1.3 Radiated power (see 3.16.3). Measure the radiated power for beams 1, 2, 3, and 4. Verify that the power in each beam does not exceed 100 milliwatts.

4.9.1.4 Velocity limits (see 3.16.4). Measure the noise output of the RTA and verify that the noise level varies by 1 dB maximum over the frequency band of 20 KHz to 40 KHz.

4.9.1.5 Altitude limits, attitude limits (see 3.16.6).

a. Measure the receiver sensitivity and verify that it is -124 dBm or less (test equivalent to 10,000 feet, land or smooth sea).

b. (Lock-on, Drop-out S/N tests on SDC) (see 4.10.8).

4.9.1.6 RTA input/output interface. The following tests are performed to demonstrate compliance with paragraph 3.16.8.

4.9.1.6.1 Beam switching, FM modulation, test discrete, IF output. Put the system into the TEST mode by setting the CDU MODE switch to TEST. Verify that after 15 seconds, a GO indication is obtained on the CDU display.

4.9.1.6.2 Antenna calibration constants. Inhibit the beam switching and set the beam cycle to Beam #1. Record the state (1 or 0) of each of the 8 antenna calibration output lines. Repeat the above for Beams 2, 3, and 4. These data provide ΔKx, ΔKy, ΔKz, Kyx, and Kxy Kxz. Compute the calibration constants from the recorded data and compare the results with the nameplate data.

4.9.1.6.3 DC power. (+5 V, ±15 V). Measure the +15 V, -15 V, and +5 V outputs of the SDC. These shall be as follows:

+15 V ±1 V; -15 V ±1 V; 5.0 V ±0.3 V

4.10 SDC, CV-3338/ASN-128( ) (see 3.17).

4.10.1 Conversion accuracy (heading, pitch, roll, TAS) (see 3.17.1.1.3). With a 26/11.8 Vac, 400 Hz synchro standard connected to the SDC heading, pitch and roll inputs, set these inputs to 180°,

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-30° and +30°, respectively. Set the TAS input to 175 KM/HR. Record the SDC data transmission to the CDU of A/D Converter Bias (J), Sin TAS (A), Cos TAS (B), Sin HDG (C), Cos HDG (D), Sin PITCH (E), Cos PITCH (F), Sin ROLL (G), and Cos ROLL (H). One hundred count samples shall be recorded for each parameter. The values of heading, pitch, roll and TAS shall be within the limits indicated below to demonstrate compliance with 3.17.1.1, 3.17.1.2, and 3.16.7.

$$HDG = \text{Tan}^{-1}\ \frac{C-J}{D-J} = 180° \pm 0.45°$$

$$PITCH = \text{Tan}^{-1}\ \frac{E-J}{F-J} = -30° \pm 0.45°$$

$$ROLL = \text{TAN}^{-1}\ \frac{G-J}{H-J} = 30° \pm 0.45°$$

$$TAS = -103.2 + (\text{Tan}^{-1}\ \frac{A-J}{B-J})* (1.03) = 175 \pm 1\text{KM/HR}$$

* This shall always be interpreted as a positive angle between 0° and 360°.

4.10.2 SDC on/off discrete (see 3.17.1.2). Apply +5 V to the SDC as the turn-on signal for the SDC. Verify that the SDC turns on by monitoring the +15 Vdc output. Reduce the turn-on voltage of 0 V, and verify that the SDC turns off. Repeat the above tests with the turn-on voltage set to 5.5 V and then to 4.5 V, and the turn-off voltage set to +0.5 V and then to -0.5 V.

4.10.2.1 Noise immunity (see 3.17.1.2.2).

a. Set the SDC ON/OFF discrete to +5.5 and superimpose 1 volt, 50 microsecond duration pulses at a 5 pps repetition rate on the line. Verify that SDC turns on. Repeat the above test with the ON/OFF discrete set to 4.5 V.

b. Set the ON/OFF discrete to 0.5 V and superimpose 1 volt, 50 microsecond pulses on the line as in the previous test. Verify that the SDC does not turn on. Repeat the above test with the ON/OFF discrete set to -0.5 V.

4.10.3 RTA, SDC radar initialization (see 3.17.2). With the CDU disconnected from the SDC, apply a +5-volt turn-on signal to the SDC and simultaneously start a timer. Examine bit 23 (table I) in any of the SDC output data words. Verify that this bit goes to "1" when the turn-on signal is applied and that it goes to "0" after an elapsed time of 18 ± 2 seconds. Repeat the above test and verify that the initialization data is as shown in table I. These tests are performed to demonstrate compliance with 3.17.2 and 3.17.2.1.

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4.10.4 Signal acquisition (see 3.17.2.2). Apply a simulated velocity to the SDC and set the S/N ratio for all four beams to 0 dB. The MEM light on the CDU shall be lit. Set the S/N ratio for Beams 1, 2 and 3 to 8 dB, and verify that the MEM light is still lit. Set the S/N ratio for Beam 4 to 8 dB and simultaneously start a timer. Stop the timer when the MEM light goes out. The elapsed time shall be 10 seconds or less.

4.10.5 SDC digital output (see 3.17.3).

4.10.5.1 Voltage level (see 3.17.3.1.1). Connect the CDU to the SDC. Turn the system on and monitor the SDC digital data line with an oscilloscope. Verify that the data line voltages are as follows:

Data voltage level "1" = +9 to +11 V.
Data voltage level "0" = -9 to -11 V.
Data voltage level synch blanking = -1 to +1 V.

4.10.5.2 Circuitry (see 3.17.3.1.2).

a. Repeat the above test with the following load on the SDC substituted for the CDU:

Resistance	-	12,000 ohms
Capacitance	-	1000 picofarads

b. Repeat the above test with the load on the data line as follows:

Resistance	-	600 ohms
Capacitance	-	30,000 picofarads

4.10.5.3 Noise immunity (see 3.17.3.1.3). Connect the CDU to the SDC. Turn on the system and superimpose one volt 50 microseconds, 10 pps rate pulses on the data line. Set the CDU mode switch to TEST and verify that after 15 seconds, a GO is displayed on the CDU.

4.10.5.4 Reverse current (see 3.17.3.1.4). With the SDC turned off, apply a variable dc voltage to the SDC digital data line. Increase the voltage until a current of 1.0 milliamperes is obtained. Reduce the voltage to zero, reverse the polarity of the voltage, and increase the current to 1.0 milliampere. Remove the variable dc source and connect the SDC to the CDU. Set the MODE switch to TEST and verify that a GO indication is obtained on the CDU display after 15 seconds.

4.10.6 Digital data waveform (see 3.17.3.2 and 3.17.3.5).

a. Load the SDC digital data line as follows:

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Resistance	-	600 ohms
Capacitance	-	30,000 picofarads

Turn the SDC on and monitor the SDC data line voltage with an oscilloscope. Measure the rise and fall times, the data bit length (BL) and the data pulse width (BL/2) (figure 5).

The rise and fall times shall be measured between the 10% and 90% voltage levels and the data pulse width shall be measured at the 50% voltage levels. The results shall be as follows:

Rise time	-	10 $\pm$ 5 microseconds
Fall time	-	10 $\pm$ 5 microseconds
Data bit Length (BL)	-	69 to 133.33 microseconds
Data pulse width	-	$\frac{BL}{2}$ $\pm$ 10%

b. Repeat the above test with the following data line load:

Resistance	-	12,000 ohms
Capacitance	-	1000 picofarads

4.10.6.1 Word structure (see 3.17.3.3). Reference tables I and II for the following tests:

a. Turn the SDC on and monitor the SDC data line voltage with an oscilloscope. Count the number of bits in any data word and verify that the number of bits in the word is 24.

b. Monitor the beam switching signal outputs from the SDC and measure the beam cycle period (time from the start of a given beam transmission to the next transmission of this same beam). Compute the digital data transmission rate as:

$$R = \frac{12}{T}$$

Where R is the transmission rate, 12 is the number of words transmitted during the beam cycle period and T is the beam cycle period. The data transmission rate shall be 93 words/second or less.

4.10.6.2 Word/bit constants (see 3.17.3.3.1). Reference tables I and II for the following tests. The required SDC digital data output shall be displayed (either on a scope or by illuminating lamps corresponding to each of the data bits) and examined for conformance to the specified requirements. The RTA, SDC, and CDU shall be interconnected for these tests.

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a. Memory. Adjust the S/N ratio for the simulated velocity input to the SDC for each of the beams to zero. Examine bit 22 in each of the 4 SDC beam velocity outputs and verify that the bits are all "0". Increase the S/N ratio in all of the beams to 8. Reexamine bit 22 in each of the beam velocity data words and verify that the bits are all "1".

b. Test. Set the MODE switch on the CDU to TEST and examine bit 23 in each of the SDC output data words. Verify that the bits are all "1". Set the MODE switch on the CDU to UTM. Reexamine bit 23 in each of the SDC output data words. Verify that the bits are all "0".

c. Power supply fail. Apply an overload to the +15 volt output of the SDC and examine bit 22 in the Sin HDG data word. Verify that this bit is a "1". Remove the overload; reexamine bit 22 in the Sin HDG data word and verify that this bit is a "0".

d. True airspeed availability. Place a short between pins m and n of connector (j) on the SDC. Examine bit 22 in the Sin TAS data word, and verify that this bit is a "0". Remove the short; reexamine bit 22, and verify that this bit is a "1".

4.10.6.3 Word Synchronization (see 3.17.3.3.2). Display at least two SDC digital output data words on an oscilloscope and count the number of bit cycles between the data words. A gap of at least 4 cycles shall exist between data words (see figure 5).

4.10.6.4 Data cycle and data rates (see 3.17.3.4). Using figure 2 as a guide, monitor each of the data parameters transmitted to the CDU for a period of 10 seconds to establish the data transmission rate for each parameter. The transmission rates shall be as follows:

Parameter	Transmission rate
B1, B2, B3, B4 Sin H, Cos H	7 - 7.5/sec
Sin R, Cos R, Sin P Cos P, Sin TAS, Cos TAS, ΔKx, ΔKy, ΔKzKyx, Kxy Kxz, Bias	3.5 - 3.75/sec

4.10.7 Velocity limits; velocity accuracy (see 3.16.5). Apply a simulated velocity as follows to the SDC IF input:

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Vx = +350 knots (+650 km/hr)
Vy = +100 knots (+185 km/hr)
Vz = +5000 ft/min (+1500 meters/minute)

Set the pitch, roll, and heading inputs to the SDC to zero. Record the Vx, Vy, and Vz outputs from the CDU, and compute the mean value of each velocity component. Repeat the above test with the following velocity input to the SDC:

Vx = -350 knots (-650 km/hr)
Vy = -100 knots (-185 km/hr)
Vz = -5000 ft/min (-1500 meters/minute)

The 1 sigma error in velocity outputs shall be less than:

	Land	Sea*
Vx	0.25% Vt + 0.1 Kt(0.185 km/hr)	0.35% Vt + 0.1 Kt (0.185 km/hr)
Vy	0.25% Vt + 0.1 Kt(0.185 km/hr)	0.35% Vt + 0.1 Kt (0.185 km/hr)
Vz	0.1% Vt + 0.05 Kt(0.093 km/hr)	0.1% Vt + 0.05 Kt (0.093 km/hr)

* Notes:

1. $Vt = \sqrt{Vx^2 + Vy^2 + Vz^2}$
2. Exclusive of water motion

Where Vt is the magnitude of the total velocity vector input to the SDC.

Use data collected during flight test (4.10.7.5 and 4.10.7.6) to evaluate/interpolate the system velocity error.

4.10.7.1 Navigation Accuracy Tests (see 3.16.5.1). Navigation flight tests shall be performed in accordance with a flight test plan submitted by the contractor and the tests specified herein in order to show compliance with the performance and accuracy requirements of section three (3), paragraphs 3.16.5 and 3.16.5.1, when inputs of attitude and velocity meet the limits and accuracy described in 3.16.4, 3.16.6, 3.16.7, and 3.17 through 3.17.1.1.3.

4.10.7.2 Installation. The DNS shall be installed in the aircraft in accordance with AV-IT5800-0001( ).

4.10.7.3 Engineering Preflight. Engineering preflight tests on the installed DNS shall be performed in accordance with AV-IT5800-0001( ).

4.10.7.4 Test Course. Navigation accuracy shall be determined by flying over a surveyed course. There shall be at least six (6) acceptable flight legs, of distances traveled of not less than 10 nautical miles each.

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4.10.7.5 Flight Test. Navigation accuracy tests shall be performed to show compliance with the performance and accuracy requirements of 3.16.5 and 3.16.5.1. The navigation accuracy shall be flown according to engineering flight tests as outlined in AV-IT5800-0001( ), using a surveyed course.

4.10.7.6 Data Reduction And Analysis. All the data collected from at least six acceptance flights (flight legs) shall be included in calculating the system accuracies. Actual distance flown, bearing, along track error, and cross track error shall be calculated according to requirements of AV-IT5800-0001( ). Use ASCC AIR STD 53/13( ), when the radial position errors as a function of distance are required.

4.10.8 Altitude limits (see 3.16.6 and 3.16.7). Apply a simulated velocity as follows to the SDC IF input:

Vx = +350 knots (+650 km/hr)
Vy = +100 knots (+185 km/hr)
Vz = +5000 ft/min (+1500 meters/minute)

Set the S/N ratio in each of the beams to 0 dB. The MEM light shall be lit. Then, set the S/N ratios for all four beams to 8 dB (simulating 10,000 feet, land or sea, worst condition) and verify that the MEM light goes out. Reduce the S/N ratio on all four beams to 0 dB. Verify that the MEM light comes on.

4.11 CDU, CP-1252/ASN-128( ) (see 3.18).

4.11.1 Area of operation (see 3.18.1.1). Enter the following coordinate data into the CDU in UTM using each of the storage positions 0 through 9 and B. After entering each set of coordinates, set the CDU MODE switch to LAT/LONG and verify that the correct Latitude/Longitude display is obtained. This test is performed to demonstrate compliance with the area of operation requirements of 3.18.1.1.

a. Spheroid - IN0 (International)

Enter	Display
40X EU 3499 2911	N 83° 59.9' E 059° 59.9'
40C ES 5811 1684	S 79° 59.9' E 059° 59.9'

b. Spheroid - CL6 (Clarke 1866)

Enter	Display
17X ND 3499 2871	N 83° 59.9' W 078° 00.1'
17T KD 7634 2025	N 47° 59.9' W 083° 59.9'

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c. Spheroid - CL0 (Clarke 1880)

Enter	Display
31Q AS 9493 5707	N 23° 59.9' E 000° 00.1'
31H GH 6328 9024	S 38° 00.0' E 005° 59.9'

d. Spheroid - BE0 (Bessel)

Enter	Display
53U PN 8695 0931	N 55° 59.9' E 137° 59.9'
49L AT 7291 2715	S 11° 30.0' E 108° 00.1'

e. Spheroid - AU0 (Australian National)

Enter	Display
55F FN 8698 9003	S 55° 59.9' E 149° 59.9'
48K YM 1292 1935	S 17° 00.0' E 107° 00.0'

f. Spheroid - EV0 (Everest)

Enter	Display
47P QN 4994 5003	N 11° 18.0' E 101° 17.4'
43S GJ 7361 7681	N 35° 00.0' E 077° 59.9'

4.11.1.1 Navigation Test (see Area of Operation 3.18.1.1). Set the velocity and attitude inputs to the CDU as follows:

Velocity -	Vx = 162.5 km/hr
	Vy = 0 km/hr
	Vz = 22.8 km/hr
Heading =	+030°
Pitch =	+10°
Roll =	+20°

Enter the following coordinate data into the CDU:

SPHEROID: IN0

DEST	COORDINATES	VARIATION
H	31T BE 4389 3214	E 000.0
4	31T BF 4959 9869	W 010.0

Note: Clear All Other Variations.

Operate the equipment for five minutes and terminate the run by depressing KYBD and simultaneously setting the velocity input to the SDC to zero. The present position coordinates shall be:

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31T BE 5047 (±15) 4413 (±25)

This test is performed to demonstrate compliance with the operational accuracy requirements of 3.18.1.1.

4.11.1.2 Velocity Test (SDC) (see 3.17.3.6). Set the velocity and attitude inputs to the SDC as follows:

S/N for all beams	=	9 dB
Velocity	=	Vx = 650 km/hr
		Vy = 185 km/hr
		Vz = 1500 meters/minutes
Heading	=	+030°
Pitch	=	+10°
Roll	=	+20°

Entering the following coordinate data into the CDU:

SPHEROID: IN0

DEST	COORDINATES	VARIATION
H	31T BE 4389 3214	E 000.0
4	31T BF 4959 9869	W 010.0

Note: Clear All Other Variations.

Operate the equipment for five minutes and terminate the run by depressing KYBD and simultaneously setting the velocity input to the SDC to zero. The present position coordinates shall be

31T BE 8240 (±75) 7386 (±82)

4.11.2 Present position updating (see 3.18.1.2).

4.11.2.1 Automatic updating (see 3.18.1.2.1). Enter the following coordinates into destination B:

N 30° 00.0' E 150° 00.0'

Set the velocity input to the CDU to zero and set the FLY-TO-DEST thumbwheel to B. Set the DISPLAY switch to DIST/BRG/TIME. Depress KYBD and then depress ENT. Set the DISPLAY switch to PP and examine the CDU data display. The coordinates displayed shall be identical to those entered above.

4.11.2.2 Manual updating (see 3.18.1.2.2). Set the velocity input to the CDU to zero. Enter the following present position coordinates:

N 30° 00.0' E 150° 00.0

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Set the DISPLAY switch to PP. The displayed coordinates shall be identical to those entered above. Depress KYBD and enter the following coordinates:

S 10° 00.0' W 100° 00.0'

The display of present position shall now be the same as the coordinates just entered.

4.11.3 Checkpoints (see 3.18.1.3). Set the velocity input to the CDU to zero. Enter the following coordinates into the CDU.

SPHEROID - CL0

DEST	COORDINATES
P	31N EM 0000 0000
1, 3, 5, 7, 9	31N EL 0000 9000
0, 2, 4, 6, 8, H	31N DM 6536 2000

Set the DISPLAY switch to DIST/BRG/TIME and move the FLY-TO-DEST thumbwheel from 0 through 9 and B and verify that the CDU display is as follows:

FLY-TO-DEST	DIST/BRG DISPLAY	
1, 3, 5, 7, 9	010.0	180
0, 2, 4, 6, 8, H	040.0	300

4.11.4 Target storage (see 3.18.1.4). Depress the TGT STOR key and verify that the TGT STOR display advances by one digit and cycles from 6 through 9 and back to 6. Repeat the above test inserting and displaying one set of UTM coordinates from 4.11.3 into each of the target storage registers.

4.11.5 Ground track-ground speed. During the test of 4.11.15, rotate the CDU function switch to GS/TK. Verify that the ground track and ground speed are computed and displayed to within ±1 degree and ±1 km/hr and demonstrate compliance with 3.18.1.5.

4.11.6 Range and bearing (see 3.18.1.6). This is checked in the CHECKPOINTS test (4.11.3).

4.11.7 Time en-route (see 3.18.1.7). Set the velocity input to the CDU to zero and enter the following coordinates:

SPHEROIDS = CL0

DEST	COORDINATES
P	31N EM 0000 0000
1	31N DL 7500 5670

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Set the DISPLAY switch to DIST/BRG/TIME. The display shall show

050.2 209
*

Set the MODE switch to BU and enter GS = 10 KM/HR, TK = 209. The time display shall change from an * to 301 and start to decrease at a rate of change equal to the remaining distance to go and the test velocities used.

4.11.8 Magnetic variation (see 3.18.1.8). This is checked in the navigation accuracy test (4.11.1.1).

4.11.9 Self-test (see 3.18.1.9). The following tests are performed to demonstrate compliance with the requirements of paragraph 3.18.9.1.

4.11.9.1 No malfunctions. With the CDU and RTA connected to the SDC, turn the MODE switch to TEST. Verify that after 15 seconds, the CDU displays GO.

4.11.9.2 CDU malfunction. Generate a CDU malfunction. The MAL light shall come on. Set the MODE switch to TEST. The failure code shall show

C X X X X X X.

4.11.9.3 SDC malfunction. Generate a SDC malfunction. Turn the mode switch to TEST. After 15 seconds, the MAL light shall come on and the failure code shall indicate

S X X X X X X.

4.11.9.4 RTA malfunction. Generate a RTA malfunction. Turn the mode switch to TEST. After 15 seconds, the MAL light shall come on, and the failure code shall indicate

R X X X X X X.

4.11.9.5 Battery. This test is performed to demonstrate compliance with 3.21.1. With primary power applied, turn the CDU MODE SELECTOR switch to OFF and remove the CDU battery. Wait at least two (2) minutes and then turn the MODE switch to LAMP test. The code EN shall be displayed. Depress KYBD and verify that the EN display is cleared.

4.11.10 Backup mode (see 3.18.1.10). Set the velocity input to the CDU as follows:

Vx = 93.5 knots (173 km/hr)
Vy = 54 knots (100 km/hr)
Vz = 0 ft/min

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Set the DISPLAY switch on the CDU to GS/TK. The display shall show

200±3 30±2

Set two of the beam velocities to memory. After five seconds, the MEM light shall come on. The GS/TK display shall show

200±3 30±2

4.11.11 Controls, switch lamps, alphanumeric displays (see 3.18.2).

4.11.11.1 Mode selector (see 3.18.2.1.1).

4.11.11.1.1 Lamp test. Set the MODE switch to LAMP test. Verify that the MAL, MEM, KYBD, TGT STOR, CLR, ENT, all the keyboard keys, and all the pinlite display sequents are illuminated.

4.11.11.1.2 Other positions. The functions of all other MODE SELECTOR positions are verified as part of the previous tests in 4.11.1, 4.11.9 and 4.11.10.

4.11.11.2 Display selector. The following tests are performed to demonstrate compliance with 3.18.2.1.2.

4.11.11.2.1 Wind speed/direction. Turn the MODE switch to BU. Set the Display switch to WIND SP/DIR. Enter

WIND SP = 020 DIR = 135 ± 1

Verify that the CDU accepts and displays the entry.

4.11.11.2.2 GS/TK. Turn the MODE switch to BU. Set the DISPLAY switch to GS/TK. Enter GS = 300, TK = 030. Verify that the CDU accepts and displays the entry.

4.11.11.2.3 XTK/TKE. Set the velocity input to the CDU to zero. Enter the following coordinates into the CDU:

SPHEROID: IN0

DEST	COORDINATES
O	18N WF 5000 2000
P	18N WF 5000 0005

Set the FLY-TO-DEST thumbwheel to "0". Perform a manual position UPDATE using the following coordinates:

18N WF 5050 0091

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Set the DISPLAY switch to XTK/TKE. The CDU display for XTK shall show L000.5. The display for TKE may be either R or L with an angle of anywhere between 0 and 360°.

4.11.11.2.4 Other positions. The functions of all other positions are verified as part of the previous tests in 4.11.2 and 4.11.4 through 4.11.8.

4.11.11.3 Destination display selector thumbwheel (DEST DISP) (see 3.18.2.1.3). The DEST DISP functions are verified in 4.11.3.

4.11.11.4 FLY-TO-DESTINATION selector thumbwheel (FLY-TO-DEST) (see 3.18.2.1.4). The FLY-TO-DEST functions are verified in 4.11.3.

4.11.11.5 Data entry keyboard (see 3.18.2.2). The data entry keyboard is functionally verified throughout the test of 4.11.

4.11.11.6 Alphanumeric display (see 3.18.2.3).

4.11.11.6.1 UTM (see 3.18.2.3.1). The alphanumeric UTM readout is verified in 4.11.1.

4.11.11.6.2 Latitude/Longitude (see 3.18.2.3.2). The Latitude/Longitude alphanumeric readout is verified in 4.11.1.

4.11.11.7 DIM control (see 3.18.2.3.3). Rotate the DIM control clockwise and counterclockwise and verify that the brightness of the alphanumeric display goes from full brilliance through DIM to full off. Test all other CDU externally controlled dimming in the same manner.

4.11.11.8 Status lamps (see 3.18.2.4). The status lamp MAL was checked in 4.11.9.2; MEM was checked in 4.10.8.

4.11.12 CDU digital inputs (see 3.19). CDU digital parameters are verified in 4.10.5.3, SDC digital data output.

4.11.13 CDU digital output (see 3.20).

4.11.13.1 ON/OFF discrete (see 3.20.1). Load the ON/OFF discrete from the CDU with 6000 ohms paralleled with 30,000 picofarads. With the CDU mode switch in the OFF position, measure the ON/OFF discrete voltage. It shall be 0 $\pm$ 0.5 V. Turn the MODE switch to UTM and measure the ON/OFF DISCRETE voltage. It shall be 5 $\pm$ 0.5 V.

4.11.13.2 Circuitry (see 3.20.1.1). Repeat the tests of 4.11.13.1 with a load of 12,000 ohms paralleled with 1000 picofarads.

4.11.13.3 Reverse current (see 3.20.1.3). With the CDU turned OFF and disconnected from the SDC, apply a reverse current of 1.0

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milliampere for 10 seconds to the ON/OFF discrete connector terminals of the CDU. At the end of this period, connect the CDU to the SDC, set the MODE switch to TEST and verify that a GO display is obtained after 15 seconds.

4.11.13.4 Noise immunity (see 3.20.1.2). With the CDU connected to the SDC, connect a pulse generator in series with the ON/OFF discrete from the CDU to the SDC and apply pulses of 1 volt amplitude and 50 microseconds duration at a rate of 10 pps. Turn the MODE switch to TEST and verify that a GO display is obtained after 15 seconds.

4.11.14 CDU auxiliary digital outputs (see 3.20.2). Connect the CDU to the SDC.

4.11.14.1 Circuitry (see 3.20.2.2 and 3.20.2.3).

a. Load the CDU Aux Output Data line as follows:

Resistance	=	600 ohms
Capacitance	=	30,000 picofarads

Turn the CDU mode switch to UTM and monitor the data line with an oscilloscope. Verify that the data line voltages are as follows:

Data voltage level "1"	=	+9 to +11 V
Data voltage level "0"	=	-9 to -11 V
Data voltage level squelch blanking	=	-1 V to +1 V

b. Repeat the above test with the load on the data line as follows:

Resistance	=	12,000 ohms
Capacitance	=	1000 picofarads

4.11.14.1.1 Noise immunity (see 3.20.2.3.1). Load the data line as follows:

Resistance	=	600 ohms
Capacitance	=	30,000 picofarads

Superimpose 50 microvolt-second impulses (50 microsecond duration, 1 Volt amplitude, 10 pps repetition rate) on the data line for 1 minute. At the completion of this period, remove the pulse source. Set the MODE switch to TEST and verify that a GO indication is obtained after 15 seconds.

4.11.14.1.2 Reverse current (see 3.20.2.3.2). With the CDU turned off, apply a variable dc voltage to the CDU Aux data line. Increase the voltage until a current of 1 milliampere is obtained. Reduce the voltage to zero, reverse the polarity of the voltage, and

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increase the current to 1.0 milliampere. Remove the variable dc source. Set the MODE switch to TEST and verify that a GO indication is obtained on the display after 15 seconds.

4.11.14.2 Digital data waveform (see 3.20.2.3.3). Figure 5 shall be used as a reference for the following tests.

a. Load the CDU data line as follows:

Resistance	-	600 ohms
Capacitance	-	30,000 picofarads.

Turn the MODE switch to UTM and monitor the CDU Aux data line with an oscilloscope. Measure the rise and fall times, the data bit length (BL) and the data pulse width (BL/2). The rise and fall times shall be measured between the 10% and 90% voltage levels and the data pulse width shall be measured at the 50% voltage levels. The results shall be as follows:

Rise time	=	10 $\pm$ 5 microseconds
Fall time	=	10 $\pm$ 5 microseconds
Data bit length (BL)	=	69 to 133.33 microseconds
Pulse width	=	$\frac{BL}{2}$ $\pm$ 10%

b. Repeat the above test with the following data line load:

Resistance	=	12,000 ohms
Capacitance	=	1000 picofarads

4.11.14.3 Word structure (see 3.20.2.4). Turn the MODE switch to UTM and monitor the CDU AUX data line with an oscilloscope. Count the number of data bits in any data word and verify that the number of bits in the data word is 32.

4.11.14.4 Standard language (see 3.20.2.5). Display UTM position (Northing) on an oscilloscope and examine bit 1 (least significant). This bit shall be a "0". Display zone pitch on an oscilloscope and examine bit 1. This bit shall be a "1".

4.11.14.5 Binary coded decimal (BCD) and binary output (see 3.20.2.5.1). Verify that the addresses of the BCD and BINARY outputs are in accordance with table III (BINARY) and table IV (BCD).

4.11.14.6 Code assignment for sign and status designation (see 3.20.2.5.2). Verify that the sign and status bits in each of the AUX output words are in accordance with figure 8 under conditions of normal operation, system failure and BACKUP operation.

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4.11.14.7 Word synchronization (see 3.20.2.6.1). Display at least two AUX output words on an oscilloscope and verify that the number of bit cycles between the data words is at least four.

4.11.14.8 Bit synchronization (see 3.20.2.6.2). Bit synchronization is verified during the tests of 4.11.14.3.

4.11.15 Navigation Bench Test (see 3.18.1.11). The following tests are performed after setting the Bench Test Discrete input to the CDU to a "1" and disconnecting the DVRS digital data input line to the CDU.

4.11.15.1 Ground track-ground speed (see 3.18.1.5). Set the CDU Mode Switch to UTM and the DISPLAY Switch to GS/TK and verify that the display reads:

160 ±3 008 ±2

4.11.15.2 Range and bearing, XTK/TKE (see 3.18.1.6 and 3.18.2.1.2.b).

a. Enter the following data into the CDU:

SPHEROID/VARIATION

DEST	SPHEROID	VARIATION
P	IN0	E 000.0

COORDINATES

DEST	COORDINATES
O	31T EF 1554 9680
H	31T EF 0000 3884

b. Set the coordinates of percent position to those of destination 0 and set the FLY-TO-DEST thumbwheel to H.

c. At any time between 1 and 2 minutes after initiating present position and selecting the FLY-TO-DEST, examine the DISTANCE, BEARING displays. These shall be as follows:

DIST	056.0 ±3.0	XTK L	000.5 ±0.3
BEARING	014 ±1	TKE L	007 ±1

d. Five minutes after the start of the run, examine present position coordinates. These shall be as follows:

31T EF 0188 (±10) 5214 (±30)

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4.12 Primary power (see 3.21)

4.12.1 Input power (see 3.21). The unit(s) shall be operated under the standard conditions described in 4.7. Measure the power input current with the CDU MODE switch in test and verify that the highest current drain recorded during test mode does not exceed 100 watts. Simulate an overload condition at the SDC power supply output. The MAL lamp shall illuminate. Remove the overload. Turn the system off and then back on and perform the tests of 4.11.9.1.

4.12.1.1 Normal Voltage. Set the primary supply voltage to +22.5 Vdc and perform the test described in 4.11.9.1. Repeat this test with the primary supply voltage set to +30.0 Vdc.

4.12.1.2 Transient susceptibility. Under the standard conditions described in 4.7, apply an 80 volt pulse for 0.2 seconds to the power input. Apply each positive pulse three (3) times within one (1) minute as specified in MIL-STD-704, category B, and perform the test described in 4.11.9.1. These tests shall be made as indicated to demonstrate compliance with 3.21.

4.12.2 Battery power. Battery power (failure) is tested in 4.11.9.5 to demonstrate compliance with 3.21.1.

4.12.3 Advisory lamp power (see 3.21.2). Set the advisory lamp power source to +28 Vdc and measure the advisory lamp power drain with the MEM and MAL lights both on. The power shall not exceed 8 watts.

4.12.4 Panel lamp power (see 3.21.3). Set the panel lamp power source to 5 V and depress the KYBD switch-lamp. The panel lamp power drain shall not exceed 8 watts.

4.13 Visual and mechanical inspection (see 3.24 and 3.25). Assembled unit(s) shall be examined for defects listed in MIL-STD-252. Prior to assembly into the unit(s), printed circuit boards shall be examined for compliance with MIL-P-55110 and the equipment drawings. Assemblies, modules and sections shall be examined for compliance with the equipment drawings and MIL-STD-252. In addition, an inspection shall be performed to verify compliance with those portions of system safety engineering which can be determined visually. The requirement of 3.24 and 3.25 shall be met.

4.14 Interchangeability (see 3.23). The mechanical dimensions and electrical performance on parts will be physically and electrically interchangeable between the unit(s) mechanical dimensions and electrical performance required by the gage and equipment drawings shall be measured to determine conformance with physical and functional interchangeability requirements of 3.23. When a listed dimension or performance requirement is not within specified limits, it shall be considered a major defect 1/. The following dimensions or performance shall be gaged or measured prior

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to assembly into the units:

a. Mechanical (dimensions). The gage points covered by the drawings listed on the applicable gage lists specified in the solicitation/contract.

b. Electrical performance (modules). Modules shall be tested to determine compliance with the performance requirements specified in the applicable gage lists specified in the solicitation/contract.

1/ This test will consist of interchanging one set of LRU cards and performing the group A electrical tests. In addition, gages will be used as required.

4.15 Weight. The unit(s) shall be weighed to determine compliance with 3.5.

4.16 Elapsed time indicator. The ETI shall be tested in accordance with sampling plan A in 4.4.2.1 of MIL-M-7793.

4.17 Luminance/particle impact noise detection. Lamps for testing selected from production lots in accordance with MIL-STD-105 shall be subjected to the following:

a. Conduct luminance performance testing of the total display output with all segments illuminated.

b. Pre-condition lamps at 80 ± 2 °C for 1/2 hour.

c. Plug lamps into sockets, mount on vibration table, and vibrate per method 2020, condition B, of MIL-STD-883 using 50 Hz, 10 G's, and 0.06" peak-to-peak amplitude. Do not perform step 3.31, Pretest Shock, in MIL-STD-883; perform Co-Test Shock Application, paragraph 3.3.4 of method 2020.1 in MIL-STD-883.

d. Repeat step a above and reject any lot, based on an AQL of 2.5, if luminance is lessened or any burst noise, exclusive of background noise or noise induced by shock blow, is detected.

5. PACKAGING

5.1 Packaging requirement. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

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6.1 Intended use. The Navigational Set, Doppler AN/ASN-128( ) in conjunction with external inputs from a magnetic heading reference and attitude stabilization data, computes positional data for navigation in UTM coordinates and provides present position, range to destination, three dimensional velocity and other steering information for external display of hover velocities or navigation steering information. The AN/ASN-128( ) can be installed in fixed or rotary-wing aircraft. In some aircraft it is intended to use the entire Navigational Set; in other aircraft it is intended to use Doppler Radar Velocity Sensor (DRVS) only and utilize the computer and control-display unit of the LORAN C/D Navigational Set, AN/ARN-114( ).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. First Article Inspection. Several samples of each item cited in Section 1 are generally required so that lengthy environmental and reliability tests can be completed on specified sample units while complete performance measurements can be made on other sample units (see 3.3).

d. Marking and shipping sample.

e. Place of final inspection.

f. Submission of the statement of treatment.

g. Technical literature required.

h. Quantity of spare parts required.

6.3 Group C inspection. Approval to ship unit(s) on order may be withheld, at the discretion of the Government, pending decision from the procuring activity, until compliance with 4.5.3 is complete.

6.4 Nomenclature. The parenthesis in the nomenclature AN/ASN-128( ) shall be deleted or replaced by a letter identifying the particular design, for example AN/ASN-128(W). The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.5 Dummy load. For purposes of this specification the term dummy load shall describe an item that is equivalent in size, weight

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and center of gravity to the units on order. The contractor shall be required to submit the dummy load for Government approval prior to its use.

6.6 Government-furnished equipment. The contracting officer should arrange to furnish the equipment listed in 3.1.1 (battery).

6.7 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited to severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification by the Government will be adjusted to make maximum utilization of the contractors quality control system and quality history of the product.

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.9 Subject term (keyword) listing.

Doppler Navigation System (DNS)
Receiver Transmitter Antenna (RTA)
Signal Data Converter (SDC)
Computer Display Unit (CDU)

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Custodian:
Army-CR

Preparing Activity:
Army CR
Project No. 5841-A068

Review Activities:
Army-AV

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-N-49098A(CR)	2. DOCUMENT DATE (YYMMDD) 90/06/05

3. DOCUMENT TITLE
Navigational Set, Doppler,AN/ASM-128( ) (XE-2)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

—

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME U.S. ARMY CECOM	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code) AMSEL-ED-TO FORT MONMOUTH, NJ 07703-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 . AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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